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Value-sensitive design in action: Designing student-centered intelligent tutoring systems with community college students and instructors

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ABSTRACT

Amidst increased interest in AI ethics in education, we delve into the design of human-centered AI intelligent tutoring systems (ITS) through Value Sensitive Design (VSD). We conducted design activities, engaging in discussions with community college students, instructors, and field experts including developers, and data scientists. The research culminated 16 designs incorporating human-in-the-loop, explainable AI, and privacy controls, emphasizing student-centricity in ITS. The study highlights the historical oversight of student and instructor values in learning platform design, demonstrating how VSD can integrate student-centered perspectives into ITS to promote trust, privacy, and agency, particularly for community college students in developmental courses. For practical insights, the study demonstrates human-centered AI principal application in ITS can fostering more ethical, effective, and student-centric educational technologies. Drawing from authentic learner and instructor experiences, this research offers perspectives on ethically and student-centered ITS design via the VSD process, expanding VSD's application into education.

1. Introduction

In many educational settings, students use learning software selected for them, typically as part of their school curriculum, without a clear understanding or control over the Artificial Intelligence (AI) processes operating in the background. To address this gap, our study employed a Value Sensitive Design (VSD) approach—an established framework for systematically accounting for human values throughout the design and development of technology. VSD integrates conceptual, empirical, and technical investigations to identify stakeholder values, explore how these values are embodied in design choices, and iteratively refine technological systems accordingly. In the context of educational AI, this means engaging with students, teachers, and other stakeholders to ensure that values such as autonomy, fairness, transparency, and trust are meaningfully incorporated into the software. Using VSD as our guiding framework, this paper presents a design-based research study that examines how human values can be integrated into educational technologies that leverage AI.

We conducted this research in a northeastern metropolitan community college, specifically focusing on developmental math courses using ALEKS (Assessment and LEarning in Knowledge Spaces)—a widely adopted AI-based Intelligent Tutoring System (ITS) but lacks human-centered design features. The students in this context have an authentic connection with the ITS, as they are reliant on the ALEKS for their academic progression, making their experiences with the system crucial for their success in advanced courses. This setting provides a unique opportunity to investigate the long-term real-life impact of ITS on students' learning experiences, particularly emphasizing the lived experiences of students and instructors.

In the broader landscape of educational technology, the integration of ITS in educational settings from primary to higher education is gaining momentum. ITS offers customized, interactive learning experiences, aligning with contemporary educational reform objectives to personalize learning (Peng et al., 2019). However, they also present ethical challenges such as potential surveillance, biases in automated assessments, and a reduction in learner autonomy (Akgun & Greenhow, 2021). Addressing these ethical concerns requires a careful examination of AI in education from a human-centered perspective.

To respond to these challenges, this study focusses on Human-centered AI (HAI), where leading tech companies have laid down frameworks to enhance AI-user interaction while tackling ethical concerns (Amershi et al., 2019; Google, n.d.). We specifically focused on

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integrating explainable AI, human-in-the-loop, and privacy control features, aiming to enhance students' trust, autonomy, and privacy.

Employing VSD, this research explores the inclusion of human values in the design of technology. VSD seeks to explore and understand diverse human perspectives and their potential influence on the prototype design of AI-infused learning software (Friedman et al., 2002). VSD process involved a comprehensive approach, examining the impact of technology on stakeholders through conceptual, empirical, and technical investigations. Our methodology was structured to provide insights into the prototype design process of ITS, involving multiple steps and diverse stakeholder engagement, including students, instructors, developers, and data scientists. The research questions driving this study are:

RQ: In what ways does the implementation of a value-sensitive design approach influence the design of ITS, incorporating considerations of AI ethics and human-centered AI?

- What specific findings emerge from the conceptual, empirical, and technical investigations?
- How do these findings manifest in the prototype design of the ITS, reflecting the integration of human values?

Our aim is to provide insights into how ITS can evolve to better align with student needs and values, contributing to a more effective and ethically sound learning experience in the context of rapidly advancing technology.

2. Related works

2.1. Ethical AI in education

Akgun and Greenhow (2021) spotlighted four pivotal ethical challenges in AI in education (AIED): privacy, surveillance, autonomy, and bias. Of these, our study focused on privacy and autonomy.

AI-driven learning platforms, by design, are data-intensive, necessitating the collection of student data. As Latham (2022) pointed out, for students to benefit from personalized learning, they have to compromise some level of privacy. Despite existing safeguards such as Family Educational Rights and Privacy Protection Act (FERPA)'s intent to protect student privacy, technological evolution has rendered FERPA's provisions inadequate (Regan & Steeves, 2019). Further complicating matters, the introduction of educational technologies into classrooms has blurred FERPA guidelines, opening the door to potential privacy violations (Stickland, 2019). As a response, states introduced a myriad of student privacy laws, leading to a disjointed regulatory landscape.

Turning to autonomy, while AI platforms promise personalized learning experiences, they risk sidelining student agency. Such platforms can sometimes dictate learning paths, removing student input in goal-setting or reviews. This predictive nature can also diminish the autonomy of both students and teachers, with the latter losing valuable student-instructor interaction opportunities when AI identifies learning gaps instead of the teacher (Amershi et al., 2019).

Despite the evident ethical challenges surrounding AIED, solutions remain elusive. While organizations like UNESCO have flagged these issues, there is a conspicuous lack of actionable strategies (Holmes, 2021). Echoing this sentiment, Floridi (2019) called for a paradigm shift from problem identification to solution creation. Hence, our research delves into solution creation by devising human-centered design strategies to tackle AI ethical challenges, focusing on ITS, widely adopted AI tool in education.

2.2. Design of human-centered AI and its application in education

To address AI ethics challenges, researchers and industry experts have delved into Human-centered AI (HAI) strategies, emphasizing effective human-AI interactions. Drawing from academic (Shneiderman, 2020; Xu, 2019) and industry (Apple, n.d.; Google, n.d.) frameworks and guidelines, this study incorporated Explainable AI and Human-in-the-loop into the ITS design. The goal of this study is to grapple with AIED ethics, thereby elevating students' learning experience and outcome through increased transparency and control in their learning journey.

2.2.1. Explainable AI and application in education

Explainable AI (XAI) offers clear, human-friendly explanations for how AI makes decisions and predictions, enhancing understanding and transparency (Floridi et al., 2020). For instance, in educational programs, XAI could elucidate how a student's performance data is used to tailor individual learning paths. Ananny and Crawford (2016) have investigated applying transparency to AI-based systems to enhance accountability with XAI. They believe that autonomy-through-openness can provide agency, competence, and comprehension of the information created by black-box AI-based systems. This openness provides critical awareness of otherwise concealed AI systems that could direct students to have control of their learning, preventing passive compliance (Berendt et al., 2020). Showing how learning data is managed by algorithms can encourage transparency and further explainability. As a result, students, not just the creators of the systems or teachers, can have better understanding over their learning process.

Existing human-computer interaction (HCI) studies have explored general XAI components that can be applied across various AI-based systems (Bansal et al., 2021; Kim et al., 2022). At the same time, studies applying XAI from educational perspectives are scarce. One exception is Conati et al. (2021), who explored how XAI integration in ITSs provided increased trust, perceived usefulness, and increased intention to use. While they found meaningful results, they indicated a major limitation that the explanations were complex and verbose. As a result, there is a need to convert the complexity of underlying AI mechanisms into simpler explanations that can be more generalizable and engaging. Likewise, we explored how empowering people to interpret and understand AI-based systems through XAI is analogous to the educational goal of providing self-efficacy, autonomy, and metacognition to students, most of whom lack an understanding of the technology behind AI algorithms.

2.2.2. Human-in-the-loop and application in education

Human-in-the-loop refers to AI systems that enable humans to provide feedback to reach hybrid intelligence for the benefit of transparency, agency, and developing robust systems (Dakuo et al., 2020; Xu et al., 2022). For instance, in an online recommendation system, users might rate or review recommended products, which the AI then uses to refine future suggestions. By integrating the human-in-the-loop features, users can provide feedback to AI systems, which contributes to personalizing user experiences and maximizing the benefit from products (Friedman et al., 2002). When the users are given the right level of control over the system through human-in-the-loop features, they are inclined to trust the system more (Friedman et al., 2002).

Human-in-the-loop feature is increasingly used in AI-based systems, and their applications in education have been investigated conceptually (Holstein et al., 2019; Seo et al., 2021); however, studies presenting implementation or empirical findings are rare. Among the few design works, Le and Jia (2022) attempted to increase learner autonomy by integrating learner control functions accompanied by a conversational agent. Specifically, they embedded a pedagogical agent and allowed the learners to customize their agent's personality–challenging or warm-hearted–to create the learning context they preferred. If the student selected the challenging parameter, there was a higher chance for a competition event to be triggered; if the warm-hearted parameter was selected, there was a higher probability for assistance events. Although the Le and Jia (2022) study was a design study that lacked empirical findings, their study is meaningful in that it illuminated the importance of learner agency and opened conversations of human-machine cooperation in using ITS with a software design. As AIED is not only design science but empirical science (Holmes et al., 2021), there is a need for ITS studies that integrate human-centered design components and empirically examine their impact on learners. Therefore, the current study explores how human-centered design components can be integrated into ITS with ethical import and tests its impact on students' trust, agency, and usability.

2.3. Value Sensitive Design: conceptual framework

Building upon previous studies that employed the Human–AI Hybrid Adaptivity in Educational Software framework (Holstein, Aleven, & Rummel, 2020a, 2020b) to capture teachers' viewpoints on AI in education, this study extends the scope to include students and other stakeholders. While studies have explored how teachers use and respond to AI-based tools, students are rarely included in designing these systems, and most studies do not clearly explain how values such as fairness, transparency, or autonomy are built into them (Alfredo et al., 2024). There is also little discussion of how these values might conflict in real classroom context. Therefore, there is a need to involve students more directly in the design of AI-powered learning systems so that they can better understand and influence how these systems make decisions that affect their learning, addressing such issues from the outset rather than after implementation (Viberg et al., 2023). In addition, there is a parallel need to involve other professional stakeholders—such as instructional designers, software developers, and AI ethics researchers—throughout the design and development process. Their participation ensures that ethical, pedagogical, and technical considerations are integrated cohesively, leading to systems that are not only value-aligned but also realistic and feasible for educational practice (Friedman & Hendry, 2019; Holmes et al., 2021).

To achieve a comprehensive understanding, a Value Sensitive Design (VSD) approach was adopted, placing human values at the forefront of the technology design process (Peng et al., 2019). VSD is especially suited to address the aforementioned gaps because it provides a structured methodology to engage to surface value tensions explicitly, and to embed values throughout the technical lifecycle of system design, rather than treating them as after-thoughts. VSD, a methodological framework, systematically addresses the impact of technology on various stakeholders. It does so by integrating conceptual, empirical, and technical investigations, ensuring a thorough, iterative, and cohesive exploration throughout the design process.

Conceptual investigations, as outlined by Friedman and Hendry (2019), delve into the core values intrinsic to technology design, highlighting potential tensions such as anonymity versus privacy. These investigations, while not empirically based, provide a robust conceptualization of human values, pinpointing foundational challenges for specific projects and creating a solid foundation for value integration. This study's conceptual lens, informed by AI ethics, human-computer interaction, and learning technologies literature, was directed at identifying key values applicable to the ITS redesign.

Empirical investigations examine the real-world use and impact of technology (Peng et al., 2019). Such investigations play a pivotal role in assessing how certain design elements may influence human values and experiences. In this context, usability maintains a nuanced relationship with human values and the ethical deployment of technology. Friedman et al. (2002) identify four possible design outcomes: (a) optimal usability and value alignment, (b) prioritizing usability over values, (c) prioritizing values over usability, and (d) ensuring that usability complements human values. In our study, the empirical lens was fixed on the prototype's design and its potential to bolster transparency and enrich the learning journey. Using value-oriented semi-structured interviews (Friedman et al., 2017), we captured diverse stakeholder viewpoints and dissected the intricate balance between usability and human values, weighing their ethical ramifications.

Technical investigations focus on identifying existing tech elements and mechanisms that support or distract human values and designing proactive systems to realize the human values identified in the conceptual investigation (Friedman et al., 2002). Although human values are considered, technology is at the center of technical investigations. In this study, the technological investigations began with analyzing the current ITS to understand how it can include human values. In conjunction with the conceptual investigations, the findings then guided the design of the value-oriented prototype (Friedman et al., 2017). Utilizing Explainable AI and human-in-the-loop strategies, the researcher crafted the initial prototype, which was then reviewed by domain experts, like developers and data scientists, to validate its technical viability.

3. Methodology

3.1. Study context: AI-based learning in community college math courses

The research unfolds within the authentic context of a northeastern community college's developmental math classrooms—a setting not typically focused on technology but on core academic competencies—-while focusing on the human values and the impact on stakeholders involved (Fittes, 2021; Friedman & Hendry, 2019). ITS like ALEKS are prevalent in preparatory courses which are critical for student advancement to college-level classes—courses that engage 60% of community college attendees (Community College Research Center, n.d.; Mojarad et al., 2018). To understand how these preparatory course experiences can be enhanced, this study probes the alignment of ITS with human values and experiences in learning, resonating with AI ethics concerns (Lim, 2021). The northeast metropolitan community college's developmental math courses utilize ALEKS and the instructors and students have been using ALEKS for substantial periods—over five years and at least one semester, respectively. Their prolonged, real-world, lived experiences of interacting with ITS sets this study apart. Unlike other research endeavors that might consider brief or experimental interactions, this investigation taps into the long-term, authentic experiences of both students and instructors. The aim is to glean insights from these in-depth feedback and reflections, shedding light on how to evolve the AI-based ITS to be more learner-centric. Regardless, the study is based on a small and demographically narrow sample from a single institution. While their lived experiences provide valuable, context-specific insights, the findings should be interpreted as exploratory and situated rather than broadly generalizable.

3.2. Participants

There were two types of participants: direct stakeholders and indirect stakeholders. Direct stakeholders were seven students and four instructors from developmental math courses that utilize ALEKS. All seven students and four instructors had at least one semester of experience using ITS. The indirect stakeholders were an instructional designer, user experience designer, software developer, data scientist, and AI ethics researcher. They provided insight into instructional design, human-centered design, and technical feasibility. Participants' demographics are shown in Table 1 and the rationale for their selections are described in 4.1.

3.3. Procedures

The design process followed the Value Sensitive Design (VSD) framework, which integrates conceptual, empirical, and technical investigations in an iterative and interdependent manner (Friedman & Hendry, 2019). The process included seven iterative steps involving both direct and indirect stakeholders. The conceptual investigation identified the relevant stakeholders and ethical components, the empirical investigation explored students' and instructors' needs, pain points, and value tensions, and the technical investigation translated

Table 1
Participants demographics: Students, instructors, and experts.

Role	Pseudonym	Profile	ALEKS or AI related experience
Direct Stakeholders			
Students	Emma	Female, White Psychology 5th Semester, Part time Returning to school after 20-year gap	2 semesters
	Tim	Male, White Computer Information Systems 6th semester, Full time Returning to school after 10-year gap	2 semesters
	Diane	Female, White Arts & sciences 4th semester, Full time Returning to school after 30-year gap	1 semester
	Lina	Female, White Public Health 2nd semester, Full time	1 semester
	Sally	Female, White Social Works 4th semester, Part time	1 semester
	Anna	Female, White, Veteran Environmental Science 4th semester, Part time Returning to school after 5-year gap,	1 semester
	Lucy	Female, White Psychology 4th Semester, Part time	2 semesters
Instructors	Amelia	20+ teaching developmental math in community college Developed college Algebra corequisite course	Founded developmental math emporium at the college Transferred math education platform to ALEKS in 2012 Provide consulting for ALEKS improvement
	Jenny	20+ teaching developmental math in community college setting Developed college Algebra corequisite course	Adopted ALEKS in since 2015 Provide consulting for ALEKS improvement
	Kate	20+ years of teaching community college level courses	Adopted ALEKS since 2016
	Bree	20+ teaching developmental math in community college	Adopted ALEKS since2016
Indirect Stakeholders			
Researcher, lawyer	Terry	20+ years practice in media, privacy, and intellectual property law	4+ years in AI ethics and worldwide AI policy research
Data scientist	Isaac	10+ years in data science specifically in dealing with voice recognition, deep learning models, traditional machine learning models	Volunteer data scientist at non-profit for developing ITS for middle school math
Software developer	Hanson	10+ years in machine learning development, system architecture design	Volunteer developer at non-profit for developing ITS for middle school math
UX designer	Jenna	5+ years UX design experience for both consumer and B2B products	Volunteer UX designer at non-profit for developing ITS for middle school math
Instructional designer	Sunny	2+ year experience in instructional design for edtech products	Curriculum developer for AI-based language learning app

these insights into human-centered AI features and evaluated the prototype's feasibility with experts.

Although these steps are presented sequentially for readability, they were not carried out as isolated phases. Each investigation informed and reshaped the others throughout the design cycle. Conceptual investigations provided propositions that guide subsequent empirical and technical work (Friedman & Hendry, 2019) were revisited as empirical and technical investigations evolved. The methodological approach of this study is represented in Fig. 1, and Section 4 details the outcomes and developments from each investigative step.

4. Results

Following VSD, this study executed seven iterative steps combining conceptual, empirical, and technical investigations. In our VSD application, empirical investigations evaluate human activity that can be observed, measured, or documented (e.g., learners' and instructors' practices, judgments, and value tensions in use), whereas technical investigations analyze how system properties support or hinder values and proactively implement value-aligned features. Accordingly, Step 2 and Step 5 are empirical, and Steps 3, 4, 6, and 7 are technical. This section details these steps, interweaving aspects of the methods employed within the process due to the interconnectedness of each phase. While the results derived from rich, long-term stakeholder experience, they should be understood as context-bound insights emerging from a single institutional setting rather than universal patterns.

4.1. *Step 1: conceptual investigation 1 – stakeholder identification*

The selection of the stakeholder was guided by the VSD framework, which emphasizes including both direct and indirect stakeholders whose experiences and design decisions influence the ethical and practical implementation of technology (Friedman & Hendry, 2019).

We selected instructors and students from this specific community college because it is known for its comprehensive use of ITS's full functionality to provide personalize learning. While many U.S. community colleges use ITS for developmental mathematics, most disable its adaptive or intelligent tutoring features and employ it primarily as a question bank to supplement lectures. In contrast, this institution was among the few that fully utilized ITS's AI capabilities to drive the entire course experience, providing a rich context for examining human–AI interaction in authentic educational settings.

We included instructors as direct stakeholders because they mediate between the ITS and students, translating algorithmic outputs into pedagogical decisions. Community college instructors often face ethical and pedagogical tensions balancing institutional performance expectations and diverse learner needs. Specifically, the institution included two instructors leading ITS implementations, possessing in-depth knowledge of the system's learning science, configuration, and student outcomes. Their expertise offered critical insights into the intersection of instructional practice, system design, and professional values such as fairness, autonomy, and student success within AI-enabled instruction.

We selected students as direct stakeholders to reflect lived experiences of using ALEKS in a high-stakes, real-world learning context—specifically, developmental mathematics courses at a community college where successful completion determines advancement to college-level study. These students represented an often-overlooked demographic in educational technology research—adult and returning

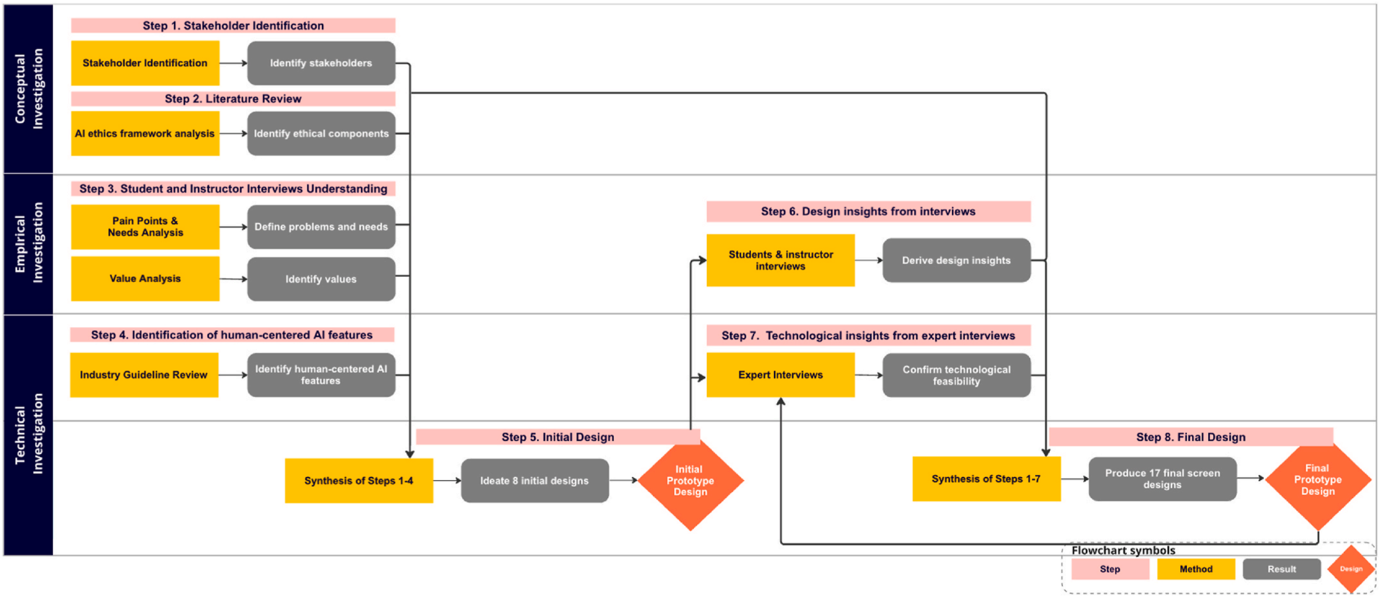

Fig. 1. Value-sensitive design process flowchart.

learners with diverse life experiences and technological fluency. We required participants to have used ALEKS for at least one semester, and the sample included both returning and new students, part- and full-time enrollment statuses, and varied majors. This diversity was essential for capturing a wide range of lived experiences and value perspectives, as students' backgrounds and learning trajectories influence how they perceive and negotiate issues such as autonomy, fairness, and trust in AI-driven systems. Their sustained engagement with ALEKS allowed for authentic reflection on how the ITS affects learning agency, trust, and motivation, providing deeper insight into how human values are experienced and expressed through ongoing interaction with educational AI.

We selected indirect stakeholders (instructional designer, UX designer, software developer, data scientist, and AI ethics researcher) to represent the professional ecosystem surrounding educational technology development. Their inclusion ensured that the study captured technical feasibility, usability design considerations, and ethical accountability from multiple disciplinary perspectives. Many had prior experience designing or evaluating educational AI systems, offering perspectives grounded in real-world edtech development and responsible AI practice.

4.2. Step 2: conceptual investigation 2 - AI ethics framework analysis

The conceptual investigation of the AI ethics framework aimed to identify core values crucial to student-centered ITS design, determining which values are significant and which are overlooked in students' use of their current ITS systems. This inquiry helped establish a foundation for selecting values to be emphasized into the ITS redesign.

We conducted an AI ethics framework analysis by delving into AI ethics literature, emphasizing core human values pertinent to our prototype's design, such as trust, agency, and transparency. We purposefully selected sources, including Friedman et al. (2005), Cheng and Fleischmann (2010), and Floridi et al. (2020), to provide complementary perspectives and ensure both theoretical and contemporary relevance: Friedman et al. (2005) established the foundational Value Sensitive Design framework that guided this study's methodological grounding; Cheng and Fleischmann (2010) offered a meta-inventory of human values serving as a comprehensive reference for identifying a broad range of potential values; and Floridi et al. (2020) articulated seven essential values for designing AI for social good, reflecting the current ethical priorities and value considerations in contemporary AI-based system development.

From these sources, we identified relevant human values both directly and indirectly related to the study through a qualitative content analysis in which two coders discussed and compared value references to determine relevance to the use of Intelligent Tutoring Systems in a community college setting.

The directly related values—ownership and property, privacy, universal usability, trust, autonomy, informed consent, accountability, helpfulness, accomplishment, honesty, self-respect, responsibility, freedom, competence, and freedom from bias—were those most implicated and foundational for guiding the empirical and technical investigations. These values shaped the development of interview protocols, the analysis of stakeholder perspectives, and the design of value-oriented prototype features. Meanwhile, the indirectly related values—identity, calmness, broad-mindedness, creativity, equality, intelligence, and justice—were considered relevant in the broader ethical context of AI in education but less directly tied to the community college ITS setting. Additionally, an exploration of the AI for Social Good Framework (Floridi et al., 2020), revealed four value-driven techniques—receiver-contextualized intervention, explanations and transparency, privacy protection and user consent, and human-friendly semanticization—which directly informed the ethical and functional dimensions of the prototype's design. At this stage, likely value associated were mapped by stakeholder role to anticipate how these would surface later in design (see Table 2):

The values identified guided the creation of interview questions and informed the student and instructor interviews analysis in Step 3. The analysis focused on the significance of these values, issues with respecting these values, and potential value tensions in students' learning with ITS. Subsequently, the insights gained from this analysis

Table 2
Stakeholder groups and anticipated associated values.

Stakeholder Group	Anticipated Associated Values
Students	Autonomy, trust, privacy, accountability, competence, responsibility, self-respect
Instructors	Accountability, fairness (freedom from bias), helpfulness, ownership and property, accomplishment, honesty
Professionals (ex. designers, developers)	Universal usability, informed consent, freedom, honesty, trust, transparency

influenced the prototype design, ensuring that each feature aimed to uphold and respect these values.

4.3. Step 3. empirical investigation 1 - student and instructor interviews for understanding target students

Step 2 focused on reflecting students' real-world use of ITS, exploring how it affected their learning, attitudes, and values when learning with ITS. This investigation was crucial for identifying how design elements in current ITS influence students' learning experiences and values to discover students' needs and pain points in learning with ITS system. In turn, we utilized these findings to inform the prototype design to better integrate values identified in Step 2. To gather lived experiences of students and instructors, we conducted 40-min interviews with students and instructors via Zoom, either individually or in focus groups, based on participant availability, to delve into ITS-related challenges and needs. To uncover key values and tensions in ITS interactions, we used the value-oriented semi-structured interview approach (Friedman et al., 2017). For instance, we asked for evaluative judgments on specific ITS functionalities by asking questions such as: "When working with ALEKS, is it alright that you do not need to actively change the way you learn the material when you face difficulties. How do you feel about it?" After interviews, we used thematic coding on students and instructors' responses and conducted member checking between the two authors to ensure reliability. Through analysis of code frequencies across participants, we confirmed that thematic coding reached theoretical saturation when no new major codes emerged after the sixth student and third instructor interviews.

4.3.1. Pain points & needs analysis

The analysis identified 14 pain points - 6 common between both groups, 4 specifics to students, and 4 to instructors - along with 7 needs - 3 common, 3 student-specific, and 1 instructor-specific. Among these pain points and user needs, those that were related to human-centered AI design were selected to be addressed in the prototype design. Participants frequently identified two main pain points: the lack of connection between math concepts caused by the AI adapting materials without showing how topics relate, and the lack of student control over review because the AI determines the learning path without learner input. They also highlighted several recurring needs, including having control over the types of help they receive from the AI system, improving communication about student learning struggles, and creating more opportunities to develop student agency and accountability.

4.3.2. Value analysis

The interview analysis revealed 10 values and 9 value tension codes, with 'trust' emerging as the most prominent value. Students and instructors expressed high trust in ALEKS' functionality and the learning science underpinning its design, attributing this trust to historical data showcasing student success rates. However, issues such as lack of transparency and unclear communication of the program's algorithms surfaced as negative factors impacting trust. The most prevalent value tension identified was 'Student agency vs. effectiveness gained though learning science behind ALEKS.' Instructors often prioritized educational effectiveness over student agency, especially for marginalized students unaware of their need for support. Conversely, four out of seven students emphasized the importance of agency in their learning process.

The identification of values, pain points, and needs informed initial prototype design in Step 4 which focused on creating a prototype optimized for usability and value alignment. By synthesizing students and instructors' voices and their lived experiences with ITS, the initial prototype design process sought to carefully balance usability of ITS with student and instructors' values. By integrating these values, the initial prototype aimed to enrich and ethically enhance the learning journey.

To clarify how empirical findings shaped early design decisions, Table 3 provides an empirical-to-technical crosswalk linking Steps 2 and 3 findings to Step 5 prototype. The table shows, for each pain point, need, value, or value tension, the specific design lever it triggered and the initial prototype screen in Appendix A where that feature appears. This mapping illustrates how empirical insights drove the technical design choices in the initial low-fidelity prototype.

Table 3
Conceptual and empirical findings mapped to initial technical design.

Pain Points Addressed	Needs Addressed	Values & Value Tensions Reflected	Initial Prototype
Step 3.1		Step 2, Step 3.2	Step 5, Appendix A
- lack understanding/transparency of own progress (C) - concept connection (C) - ALEKS & instructor's explanation difference/lack why (C) - lack understanding of topic selection (S)	- Making concepts connections (C) -repetition/deconstruction opportunities (S)	**Values:** - Trust (Friedman et al., 2006) - Helpfulness, honesty (Cheng & Fleischmann, 2010) - Explanations and transparency, human-friendly semanticization (Floridi et al., 2020) **Tensions:** Student agency vs. effectiveness gained though learning science behind ALEKS	Screen 3 – Explainable AI: Why and How Questions Are Given
- review, lack control (C) - limited pace/personalization (S) - lack understanding of real-time learning (C)	- control over types of help (C) - struggle communication/recognition (C) - agency (C)	**Values:** - Agency, Universal Usability (Friedman et al., 2006) - Helpfulness, Accountability (Cheng & Fleischmann, 2010) **Tensions:** N/A	Screen 5 – Human-in-the-loop: Asking for Student Choices in Follow-Up

Note. C = code raised by both students and instructors; S = code raised by students only; I = code raised by instructors only.

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)

- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

4.4. Step 4. technical investigation 1 - identification of human-centered AI features from industry guideline

Step 4 focused on identifying existing tech elements and mechanisms that can support implementing values identified in step 2 to the initial prototype design in step 5. To define the HAI design techniques to apply, we reviewed human-centered AI design strategies from industry sources (Amershi et al., 2019; Apple, n.d.; Google, n.d.) were reviewed. The review yielded seven human-centered AI tech elements to be integrated into the prototype. They were: (1) maximizing the educational values of AI, (2) privacy, (3) explainability, (4) revealing confidence level, (5) mental models, (6) agency from user feedback and control, and (7) providing multiple options with implementation strategies.

For education-focused AI values code, three strategies surfaced: identifying the intersection of user needs and AI strengths, balancing user control and automation, and aligning perceived and actual user value (Google). For privacy, seven strategies emerged such as understanding when users wish to maintain control (Google), returning control to the user (Google), and aiding people in controlling their information (Apple).

Transparency was addressed with codes for explainability, revealing confidence level, and mental models. For explainability, Google and Apple advised communicating benefits using simple, direct language while avoiding overly narrow or broad explanations. Revealing AI model confidence levels involved translating them into comprehensible concepts while adjusting result presentation based on confidence thresholds (Apple), For mental models, Google suggested that their inclusion in the onboarding phase would calibrate user trust, and all three companies suggested setting realistic AI model expectations.

User agency was promoted through feedback collection and addressing and by offering multiple options. Feedback-related codes included guidelines on planning for user-machine co-learning (Google), offering feedback editing opportunities (Google, Apple), adapting to the user journey (Google), communicating feedback value to users (Google), and immediate response to explicit feedback (Apple), alongside continuous updates based on evolving user interests and preferences (Google). For offering multiple options codes, Google and Apple proposed providing options - categorical or N-best alternatives, numerical options (Google), and avoiding overwhelming users with excessive, complex options by simplifying and prioritizing the most plausible ones (Apple).

The identification of the most pressing values, pain points, and needs aimed to inform initial prototype design in Step 4 which focused on creating a prototype optimized for usability and value alignment. By synthesizing students and instructors' voices and their lived experiences with ITS, the initial prototype design process sought to carefully balance usability of ITS with student and instructors' values. By integrating these implications, the initial prototype aimed to enrich and ethically enhance the learning journey.

After extracting key techniques essential for integrating human-centered AI features into the ITS design from industry reviews, we transferred them to an online board as sticky notes. This method facilitated flexible combination and integration of these techniques into the initial design in Step 5 to support values identified in Step 2.

4.5. Step 5. technical investigation 2 - initial design of low-fidelity prototype

We guided the formation of the initial design through a value-oriented prototyping process (Friedman et al., 2017), integrating findings from Steps 1 through 4. We transcribed the extracted codes from each step onto sticky notes on an online board and analyzed them using the affinity diagramming technique. This technique, traditionally employed in UX research and recently adopted in academic studies (Lisle et al., 2019), systematically categorized related data into broader conceptual categories (Scupin, 1997). The purpose of affinity diagramming was to create combination of codes from all investigations to design a low-fidelity prototype of a student-centered ITS. Affinity diagramming's experiential and intuitive nature made it ideal for prototype ideation and refinement.

We began the design process by categorizing student needs and pain points from Step 3 to determine which could be addressed together. Next, we linked these needs and pain points to the corresponding human values from Step 2. Once we identified which needs, pain points, and values should be addressed, we referred to the techniques identified in Step 4 to determine which could be applied to address these issues. The combination of these codes and techniques resulted in the creation of a low-fidelity prototype design.

Fig. 2 depicts an example of the process. First, we classified the key pain points to be addressed together: students' issues related lack of comprehension of how ALEKS operates, and understanding of the selection process for learning topics; and instructor's challenge of fostering student trust in ITS by explaining the underlying learning science. We also grouped students' and instructors' need for the capacity to verify their learning progress into this category. Then, we defined which human values were violated or needs to be supported in relation to this case: trust and autonomy from the values codes (Friedman et al., 2005), along with explanations, transparency, and human-friendly semanticization guidelines derived from the AI4SG framework (Floridi et al., 2020). Then, we selected techniques that could support these instances with the relevant human values in mind and decided to include the AI system's confidence-level reveal feature to enhance transparency. The initial design that resulted from synthesizing insights across Steps 1–4 is presented in Fig. 3 below.

The initial prototype encompassed eight distinct designs that tackle issues such as employing explainable AI to justify topic assignments to students, keeping students-in-the-loop by soliciting their perspectives on navigating challenging topics, and informing students on managing their privacy settings. The remaining designs followed a similar procedural trajectory as illustrated in Fig. 5, and these can be found in Appendix A.

4.6. Step 6. empirical investigation 2: design insights from student & instructor interviews

To conduct empirical investigations on the initial prototype from Step 5, we engaged students and instructors in 20-min individual or focus group Zoom interviews, depending on their availability. Following the Value Sensitive Design (VSD) framework, this step constituted an empirical investigation because it focused on observable human activity; specifically, participants' reactions, judgments, and evaluative feedback toward the redesigned ITS prototype. At this stage, where developing a fully functional ITS was not feasible, gathering stakeholder responses through interactive prototype sessions provided a practical and effective means of capturing how users experience and interpret value-laden design features.

These interviews aimed to understand how the initial prototype design might influence their learning experiences and values in learning with ITS. This investigation was crucial for ensuring the final prototype design to closely reflect the real-world experiences and values of students and instructors. The goal was to create a final prototype that struck a balance between usability and values, carefully considering ethical implications and incorporating stakeholder perspectives.

We used the participatory speed dating technique (Davidoff et al., 2007) to introduce participants to the newly designed human-centered intelligent tutoring systems without necessitating any technical implementation. We initially showed participants a brief video explaining the concepts of human-in-the-loop and explainable AI features to help them

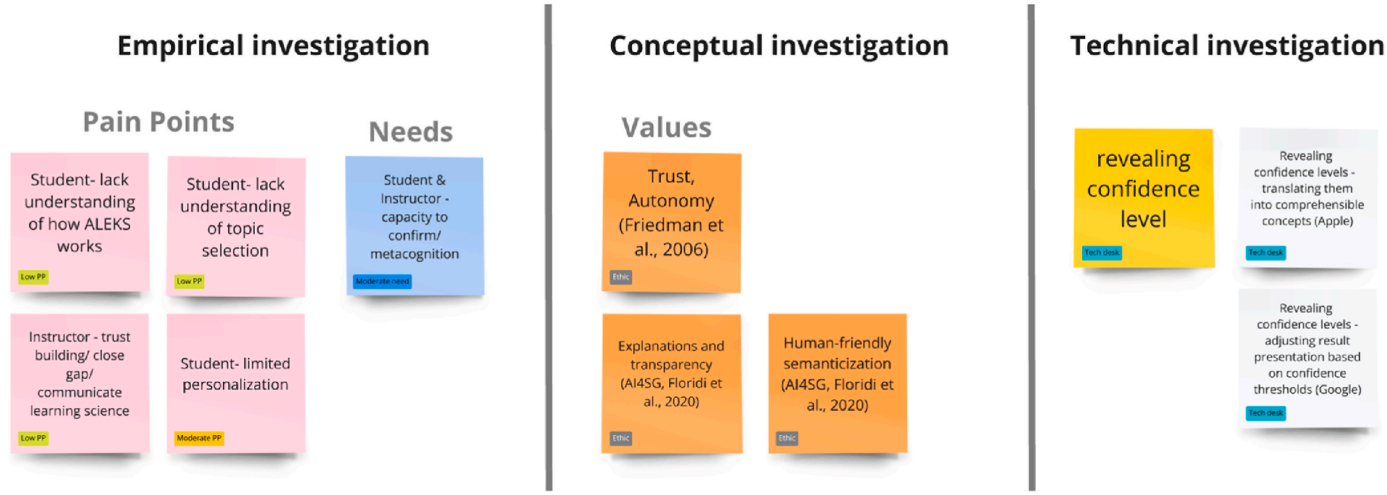

Fig. 2. Example of initial prototype design process.

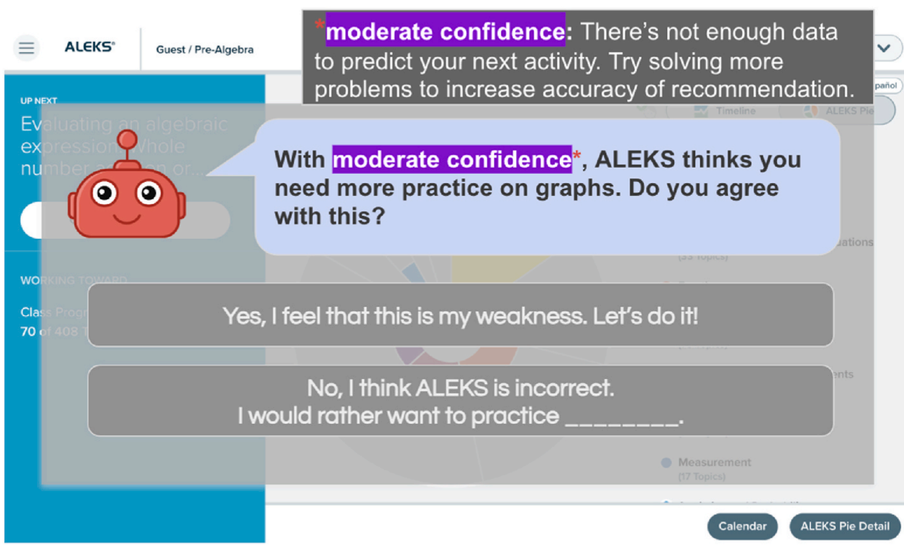

Fig. 3. Initial prototype example: Revealing confidence level of intelligent tutoring systems.

understand the design intentions. Subsequently, we presented eight initial prototype designs while their immediate feedback was recorded (Seo et al., 2021). This feedback focused on aspects such as potential usage, learning impact, possible setbacks, influence on values, and suggested enhancements. For data analysis, thematic coding method was utilized and revealed three common themes: explainable AI, human-in-the-loop, and privacy features. As a result, our analysis yielded 11 codes for explainable AI design, nine for human-in-the-loop involvement, four for privacy, and two general codes spanning multiple design aspects.

4.6.1. *Explainable AI design*

These codes explored topics such as providing options to access explanations of assessment results and the rationale behind question assignments, enabling us to identify ways to help students better understand their learning journey. Additionally, the elucidation of concept structures or the interconnections among learning concepts, such as the foundational concepts for a complicated math problem, aided students in scaffolding knowledge for application.

One advantage of explainable AI in ITS was the potential to engage students to be more diligent in working with ITS. Anna, a student, shared her personal experience of enrolling in the same developmental math course twice, the first time dropping out due to lack of understanding about how ITS worked. On her second attempt, an in-depth explanation from her professor about the ITS operation was a game-changer. She noted, “Last semester, we just jumped right in. I got all this thrown at me, and I just did not understand what was going on until it was kind of too late … I feel like a lot of people don't take that first knowledge check very seriously. They just want to guess through it. Press ‘I don't know,’ Google, which is not going to be helpful because it's not going to make everything accurate … So, I did it very, very honestly (this semester) to see what I would get without any help, just me.” Anna's anecdote underscores the importance of ensuring students understand the workings of ITS to derive the most benefit from it. However, the instructors' interviews suggested a gap in instructors' understanding and the emphasis they place on communicating the learning science behind ITS to build trust. Thus, we identified a recurring code among both students and instructors: the need for continuous communication of the learning science behind the ITS. Instructors noted that a one-time explanation of the learning science underpinning ALEKS at the semester's start was insufficient to establish student trust.

4.6.2. *Human-in-the-loop AI design*

These codes covered aspects like enabling students to control how they receive help from ITS. A student, Sally, emphasized the value of the ITS prompting to receive help, stating, "Not everybody feels comfortable reaching out and asking for help if they're struggling. Sometimes they'll just give up, log off and say, I'm done. I just can't do it tonight, or they'll drop class. It's just a lot easier to reach out when you are given options." Diane, another student, also advocated for an AI system that could prompt help when students are stuck on a topic, saying, "Sometimes I'm not good at recognizing myself. I like that instead of sitting there and trying to fumble through on my own." The students proposed numerous ways they would prefer to receive help. Emma suggested an embedded drop-down menu with additional help resources, while Diane suggested the idea of a peer mentor or the option to reach out to an instructor triggered by AI.

Furthermore, students desired for both automated and self-triggered review opportunities, with a particular emphasis on the ability to flag topics for review or repeated practice. Diane conveyed this need by stating, "I prefer to be the person to decide. Okay, I've got this piece, I understand. Let's move to the next piece instead of the machine, saying, you don't understand this piece. So, we're going to bump ahead through things, and maybe we'll come back to this." However, instructors noted the difficulties in providing student control over reviews due to the existing algorithm-driven design of ITS to provide ideal learning paths. The matter of whether students should have review control ignited debate among instructors. While Jenny and Beth highlighted the importance of fully trusting the ITS's learning path and the review opportunities it provides, Kate expressed understanding of the students' frustrations over the perceived lack of control.

The students suggested marking their confidence at the end of a practice session to convey their metacognitive understanding to the AI for a more personalized learning path. This suggestion stemmed from the recognition that students might pass topics without having full confidence in their understanding. By marking their confidence level, students might be able to better communicate their understanding to ALEKS. Additionally, Amelia and Bree, instructors, endorsed the idea of allowing students to note whether they consulted external help in completing practice questions, aiming for a more accurate assessment of their learning progress.

4.6.3. Privacy design

The privacy control design from a student's perspective generated several themes: students' desire for control over privacy with instructors or ITS companies, the need to raise awareness about privacy issues in edtech, and specific considerations for how these privacy options and messages are designed.

Views on sharing learning analytics data with instructors or institutions were divided. While four out of seven students saw no need for privacy settings, believing instructor access to their learning progress could be beneficial, three students voiced a desire for privacy settings. Emma cited anxiety support, Diane wanted a sense of control, and Tim wished for a feeling of respect. Meanwhile, all instructors expressed uncertainty over how privacy settings would operate as they felt the need to assess students' learning data to support their learning.

When it came to sharing learning analytics data with the ALEKS company, students held a variety of opinions. Sally stated, "Yes, if I was informed about it (benefits of sharing learning data for better learning experience of others), I would have no problem sharing with ITS company to help build a better system." Contrarily, Lina voiced a need for separate privacy controls, saying, "If they're (ALEKS) using my data, then this (ALEKS) should be free. I shouldn't be paying for this. If this data is used, they are literally profiting off of me." The discussions also revealed a general low level of awareness around privacy issues when using edtech products. Tim never considered the privacy implications in using ITS. He stated, "If it asks me this question (privacy setting), then I have it in my head that 'should I trust this or not?'" Conversely, Sally noted, "In this day and age, people want to know what's being shared, why, and with who naturally. So, I think this would help a lot and it allows … control how much is being shared and for what reasons, how often."

Bree, an instructor, suggested an ideal privacy setting design should provide clarity on the definition of data analytics, what data is being collected, the purpose of this collection, how it benefits learners, the control students have over their data, and all this information should be communicated in learner-friendly language. Bree emphasized that expressing that HITbot collects data to improve the student's experience on ALEKS could help alleviate fears of data sharing. Bree noted, "What is the learning data HITbot is collecting, expressing that the HITbot is collecting data to give you more control over your experience on ALEKS. This would help combat fear of sharing data."

4.6.4. General design

Two universally applicable principles were identified across all design elements: the need for the AI to send out collaborative messages in a congenial tone, and the importance of timing when AI agents engage with students. Both students and instructors deemed it significant for the AI agent to send out collaborative messages. Bree, an instructor, emphasized that convincing students that the ITS is genuinely invested in students' best interests could strengthen their trust and compliance with the system: "'I truly do have your best interest at heart, which is why we're (AI and student) doing this. I suggest you hang in there' … I think it (this message) can reduce frustration and foster understanding." A student, Lucy, also appreciated the collaborative tone, describing the AI agent as a 'Robot buddy' in the learning journey: "I'm here to help you. Let's figure out where you are.' (So, a collaborative message?) Yes, so that way, a little Robot buddy is here to help us succeed and do great." Regarding the timing of messages, participants expressed that the moment of support delivery is crucial. For instance, a student, Ally, underscored the importance of the AI agent's engagement during the learning process, not just at the end.

4.6.5. Designs that surfaced critical value tensions

Two initial designs confused students and instructors, but we believe that they revealed important value tensions and design insights for navigating conflicts between designer intention and user interpretation.

The first of these was the revealing of the AI model's confidence level, a feature intended to be part of the explainable AI functionalities. We observed that the "confidence level" concept often led to misunderstandings among instructors and students, with many confusing it with "students' confidence in learning." One student and one instructor didn't understand it, while two instructors didn't see the need to reveal confidence levels. Another student, although understanding the concept, suggested hiding detailed information about it to avoid confusing those not interested. Ultimately, only one student and one instructor comprehended its original purpose and found it useful. Bree, an instructor who understood the usage of confidence revealed commented on potential positive impact, "I think students respond well to whether it's a teacher or a system being real with them about their imperfections … It also humanizes the experience. I think there is a more user-friendly way of saying such as AI gets better and better at making this meaningful to you. You are a big part of this."

These reactions highlighted an important value tension in educational explainability: model-level transparency does not automatically translate into meaningful user understanding. Instead, we believe that explainable AI in education must be framed in ways that directly support learners' goals and reduce interpretive load, rather than presenting raw system metrics, supporting effortful learning rather than technical transparency for its own sake. For instance, we could reframe the confidence level in more humanizing terms, for instance: "the AI might be uncertain here, so let's check this together".

The second design was sharing emotional analysis with instructors. This concept proved controversial among both instructors and students. For instance, Emma, a student, noted, "It doesn't seem very appropriate to assess emotions … maybe the language could be changed because I actually think the information is very helpful to the professor." An instructor, Kate, found the information beneficial for enhancing communication in online classes: "With so much going on online, it's hard for an instructor to know exactly when a student needs assistance. I'm looking at a lot of analytics on there, how many topics they've got, but I don't see how difficult it was for them, or if they were frustrated while they did one and the other one, they found very easy. I don't get any of that kind of feedback without talking to the student. More communication is good." However, to reduce controversy, the design was alternated with different ideas. This divergence exhibited an important value tension that privacy, dignity, and autonomy may conflict with instructional benefits. This case showed that we should avoid collecting or sharing sensitive data (e.g., emotions) unless users clearly understand its purpose, can opt in, and perceive direct benefit. This aligns with educational ethics frameworks that caution against collecting or inferring sensitive learner data, such as affective states, without explicit consent or a demonstrated instructional purpose (Akgun & Greenhow, 2021; Holmes et al., 2021; Regan & Steeves, 2019).

Step 6 revealed key changes to the initial prototype. One of the most significant shifts was the integration of more student control features, such as explainable AI and human-in-the-loop options, which emerged as crucial for fostering trust and engagement. However, these change also highlighted tensions, especially around the balance between giving students autonomy and maintaining the integrity of the ITS learning path. While students valued control over their learning journey, instructors were divided on whether this could undermine the system's effectiveness. Privacy concerns also surfaced, with students expressing a desire for more control over their data, while instructors questioned the practicality of such options. We managed these tensions by refining designs to accommodate student preferences without compromising the ITS's educational integrity, such as modifying privacy settings and adjusting how feedback and explanations were delivered.

Together, these reactions highlighted broader design tensions such as transparency vs. interpretability, privacy vs. instructional insight, and student agency vs. system-guided scaffolding. These shaped how features were revised, simplified, or reframed in the final design. Accordingly, insights from Step 6 were incorporated directly into the Step 8 prototype, guiding refinements to existing features and prompting the redesign or removal of elements that risked value misalignment. To illustrate how conceptual and empirical insights informed these changes, Table 4 highlights two examples showing how findings from Steps 2, 3, and 6 translated into concrete design decisions in the final prototype in Step 8.

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)
- concept connection (C)
- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

Pain points (Step2).

- lack understanding/transparency of own progress (C)
- concept connection (C)

Table 4
Conceptual and empirical findings mapped to final technical design.

Needs, Pain Points	Reactions, Judgments, Suggestions	Values & Value Tensions Reflected	Final Prototype
Empirical Insights from Step 3.1	Empirical refinements from Step 6	Conceptual and Empirical Insights from Step 2, Step 3.2	Step 8, Appendix B
• Need (C) – setting up student expectations • Pain point (C) – assessment interpretation/ clarity • Pain point (I) – trust building/ communicate learning science • Pain point (S) – lack understanding of how ALEKS works	• (C) – communicating learning science *continuously* (not only at start of semester) • (S) – explaining concept structure/ scaffolding • (S) – explaining why student got questions/ reasoning behind path • (I) – friendly/ humanizing semantics/ wording • (S) – human-AI collaborative message	**Values:** • Trust 3 – Lack explanations, transparency, communication (Trust that is weakened by the system's lack of explanations, insufficient transparency, and poor communication about its adaptive decisions.) • Universal usability • Ownership **Tension:** • Transparency vs. Collaboration (Students preferred collaborative messages over raw model transparency, because transparency alone has little value unless it directly supports their learning.)	**E2 – Setting Expectations on Learning Science**
• Need (S) – repetition & deconstruction opportunities • Need (C) – student agency/ accountability development • Pain point (C) –review, bookmark/ memo • Pain point (C) – knowledge retention • Pain point (S) – limited pace/ personalization	• (S) – serving individual learning style • (S) – students marking confidence on mastery • (C) – student control over review • (S) – explaining how repetition helps understanding	**Values:** • Agency • Responsibility • Competence • Metacognition • Universal usability **Tension:** • Student agency vs. equity (Instructors argued that model-driven adaptivity should sometimes override student choice because lower-performing students may not recognize when they need support.)	**H4 – Bookmarking for Review**

Note. C = code raised by both students and instructors; S = code raised by students only; I = code raised by instructors only.

- ALEKS & instructor's explanation difference/lack why (C)
- lack understanding of topic selection (S)

4.7. Step 7. technical investigation 3: expert interview insights

To confirm the technical feasibility of the initial and final prototype and identify additional technological elements from a practitioner's perspective, we conducted a series of interviews with five field experts: AI ethics researcher, instructional designer, data scientist, software developer, and UX designer. The interview structure was tailored to expertise and the prototype's development stage. For instance, the AI ethics researcher offered conceptual insights into human values after reviewing the initial prototype. Similarly, the data scientist examined the technical feasibility of the features during two stages: after the initial and final prototype review. The instructional designer's feedback focused on instructional strategies and learner engagement, given after examining the first version of the final prototype. The UX designer critiqued the prototype's usability twice: after reviewing the first version of the final prototype and once more after implementing their suggested changes. Lastly, the software developer evaluated the technical feasibility of the features after the final prototype review. These 30-min expert interviews were conducted individually over Zoom. Because the five experts represented distinct areas of specialization, their feedback reflected diverse disciplinary perspectives. Rather than seeking common themes, we treated their insights as complementary evaluations, each offering a unique lens on the prototype's technical feasibility and ethical soundness. We documented key points from these interviews on an online board and used them as a design-integration checklist, allowing us to translate expert feedback directly into actionable design improvements. This approach was more effective than traditional thematic coding for synthesizing cross-disciplinary expertise.

4.7.1. AI ethics researcher

The researcher provided conceptual insights after evaluating the initial prototype. She emphasized the importance of incorporating human values into the design of intelligent tutoring systems, such as privacy, freedom, trust, autonomy, informed consent, and calmness. She stated, "It is important to make sure that humans are not judged by machines and through explanations build trust between human and machine." Furthermore, she offered practical suggestions for implementing these human-centered features. For example, she advised a more user-friendly presentation of the confidence feature. She suggested, 85% confidence level is incomprehensible to the users and providing leveling with explanations on what this means to the students in their learning context. When it came to explainable AI, she recommended explaining 'why' is usually more digestible for users, but 'how' can be more technical and thus challenging and repetitive. She therefore proposed, "combining 'why' and 'how' into one explanation. In terms of privacy control design, she advocated for transparency and clarity, stating, “Include information on what is shared, to whom? how much? how long is the information kept for, explanation of what is done … short and simple.” These practical design insights from the AI ethics researcher were incorporated into the final prototype's design, further enhancing its human-centric focus.

4.7.2. Instructional designer

Following the review of the initial and final prototypes, an instructional designer offered feedback focused on instructional strategies and learner engagement. As an ITS operates based on students' information, she emphasized the significance of informed consent as a human value and stated that transparency was critical concerning the collection and utilization of student data. In her view, ITS is best suited to support individualized learning, which requires guaranteeing learner autonomy. Thus, she also stressed the importance of autonomy as a core human value. To increase learner engagement, she recommended incorporating features seen in successful platforms like Duolingo. She proposed enabling students to establish personalized learning goals, adding a friendly character to represent the AI agent, enhancing the reward system, and incorporating a mechanism for students to give feedback to the system, particularly when the ITS does not align with their learning needs. Her feedback provided insights to enhance its instructional efficiency and learner engagement.

4.7.3. Data scientist

The data scientist scrutinized the technical feasibility of the initial and final prototypes. He emphasized the importance of privacy, universal usability, trust, and autonomy in using ITS. He acknowledged the technical viability of the initial designs, though he pointed out potential challenges with implementing features that would share students' affective states with instructors using advanced technology like facial recognition. He also raised questions about allowing students to control the sharing of their learning data with instructors. Despite the absence of technical obstacles to the feature, he expressed concerns about the possible repercussions if students decided against sharing their data, adversely affecting the overall course experience. He suggested a system that recalls students' choices like privacy settings and sets them as default for future sessions, enhancing machine learning for a personalized experience. The data scientist also voiced his belief that the explainable AI features would boost learner agency. As he saw it, students would be prompted to revisit topics, thus promoting deeper learning. Additionally, the transparency of system actions and rationales from explainable AI could foster trust in the system owing to informed choices. He highlighted that the industry traditionally prioritized system accuracy over human-centered design but believed a shift towards a human-centric approach would benefit students. Finally, the data scientist underlined the importance of having a large dataset for the prototype's features to function effectively.

4.7.4. Software developer

Following the final prototype assessment, the software developer contributed insights on technical feasibility and potential technological enhancements. He recognized the overall technical viability of the final prototype but highlighted some challenges. For instance, he raised concerns about revealing confidence levels as it would require the AI to self-diagnose its accuracy and address potential challenges in defending it. Regarding the individualized goal feature, he proposed introducing flexible goal setting by AI and effectively communicating its adaptability to students. Emphasizing the value of Explainable AI, he confirmed its feasibility for enhancing learning outcomes. He suggested that labeling questions with concepts titles could help developers discern student struggles and subsequently provide explanations. He emphasized the importance of acknowledging individual user preferences and advocated for giving users control over their involvement levels, deeming the feature technically viable. In alignment with the data scientist, he emphasized the significance of a substantial and diverse dataset to develop human-centered features, ensuring fairness and customization for different user groups. While emphasizing fairness and customization, he warned of usability challenges stemming from extensive data collection.

4.7.5. UX designer

Upon evaluating both the preliminary and the finalized prototypes, the UX designer offered suggestions for enhancing the usability of the prototype. She emphasized the importance of concise messaging, cautioning that excessive verbosity can deter users. Moreover, she suggested the adoption of two primary color themes for aesthetic consistency and recommended employing different color fonts to emphasize key messages. She also proposed an alternative location for the AI agent character to enhance user engagements and stressed the significance of enabling users to undo their prior actions. These practical design recommendations were incorporated into the final design to optimize user experience.

The expert interviews confirmed the technical feasibility and refined features. One major shift was the emphasis on explainability and human-centered design, particularly in clarifying the AI's decision-making processes and enhancing user control over privacy settings and learning paths. The tensions that arose centered around balancing transparency, student autonomy, functionality, and technical feasibility. For example, while students desired more control over their learning data, experts raised concerns about the potential drawbacks of withholding data from instructors. Additionally, the challenge of implementing explainable AI features like confidence level indicators required careful consideration of how to communicate these concepts without overwhelming students. In Step 8, the final design incorporated expert suggestions and addressed tensions by simplifying user messaging in layman's terms, enhancing student engagement in instructional decisions, offering more diverse options for receiving help, and giving students control over their privacy settings. The goal of the final prototype was to balance technical feasibility with the integration of human values, such as trust, autonomy, and privacy, into the ITS design.

4.8. Step 8. technical investigations 4: final design

We developed the final design through a value-oriented prototyping process using affinity diagramming, as in Step 5 (Friedman et al., 2017; Lisle et al., 2019). Codes and notes from Steps 1-7 were systematically organized on sticky notes, ensuring that findings from each step were incorporated into the final prototype. The process began by refining the initial designs from Step 5, integrating feedback from the student and instructor (Step 6) and expert (Step 7) interviews. Direct quotes from interviews were also considered. Moreover, additional design insights from Step 5 prompted us to revisit the codes from Steps 2-4 to identify further design opportunities. Although these new insights guided the design process, maintaining the core values from Step 2 remained the primary goal. The refinements in the final prototype reflect the value priorities of participants in this specific community college context and are not intended as universally prescriptive design requirements.

Fig. 4 illustrates an example of the process to create a new design. This design was inspired by quotes from students and instructors expressing the need for student feedback to improve ITS functionality:

> "The idea is ALEKS thinks you need this; do you think you need it? And then challenge that so that the student can see that they need it." (Amelia, Instructor)

> "I would have pointed out in that situation like 'this is not helpful' because I wasn't given the information … doesn't mean it's wrong, but you could get some feedback saying like this was put in here, for this reason … think having that interaction would really contribute to the program being more successful and more helpful." (Sally, Student)

> "Contribute feedback, or … would you like to contribute any feedback on this? I think knowing that you are contributing to the program, you'd feel empowered."(Emma, Student)

To respond to these quotes in design, we first identified insights from Step 6 that were not addressed in the initial design, such as providing feedback to ITS for enhanced personalization and communicating students' contributions through human-in-the-loop features. We then evaluated how these features could impact students' values in learning with ITS, such as ownership, autonomy, accountability, helpfulness, and self-respect (Cheng & Fleischmann, 2010; Friedman et al., 2005). Additionally, we drew human-centered AI elements from the AI for social good framework (Floridi et al., 2020), which were receiver-contextualized intervention and human-friendly semanticization.

We also examined whether this new feature could address the pain points and needs identified in Step 3, such as: students' lack of control over review opportunities; limited understanding of their progress; the need for greater personalization and transparency; and need for enhancing student agency and accountability in learning with ITS.

Finally, we referred to techniques from Step 4, such as planning for co-learning between the user and the machine (Google, n.d.), collecting feedback to improve personalization (Apple, n.d.), and communicating the value and impact of the feature to students (Google, n.d.). These combination of insights and techniques led to the creation of final design that allows students to provide feedback to ITS (see Fig. 7).

This process led to 16 final designs, organized into three categories of human-centered AI: (1) explainable AI, aimed at enhancing students' understanding of ITS functionality and their learning journey, (2) human-in-the-loop features, offering greater control and assistance options, and (3) privacy controls, which increase awareness and provide more control over learning data. Four key designs are highlighted below, with the remaining screens detailed in Appendix B.

First, Fig. 6 showcases an explainable AI design. For instance, when students are practicing fractions, ITS might detect gaps in their foundational knowledge and present related questions without explaining

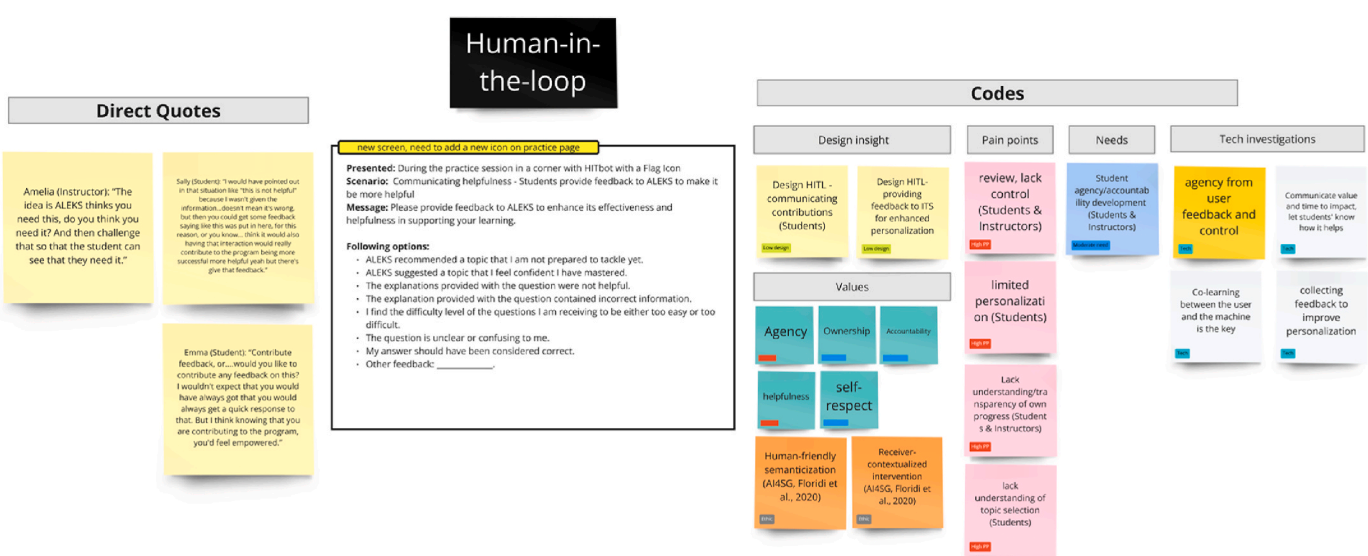

Fig. 4. Final design process example.

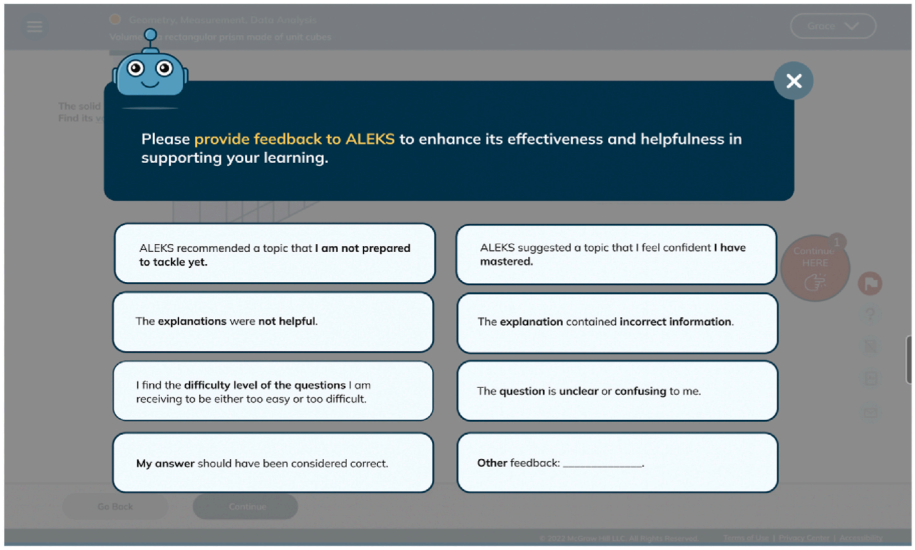

Fig. 5. Human-in-the-loop Feature: Providing Feedback to ITS to Enhance its Effectiveness and Helpfulness.

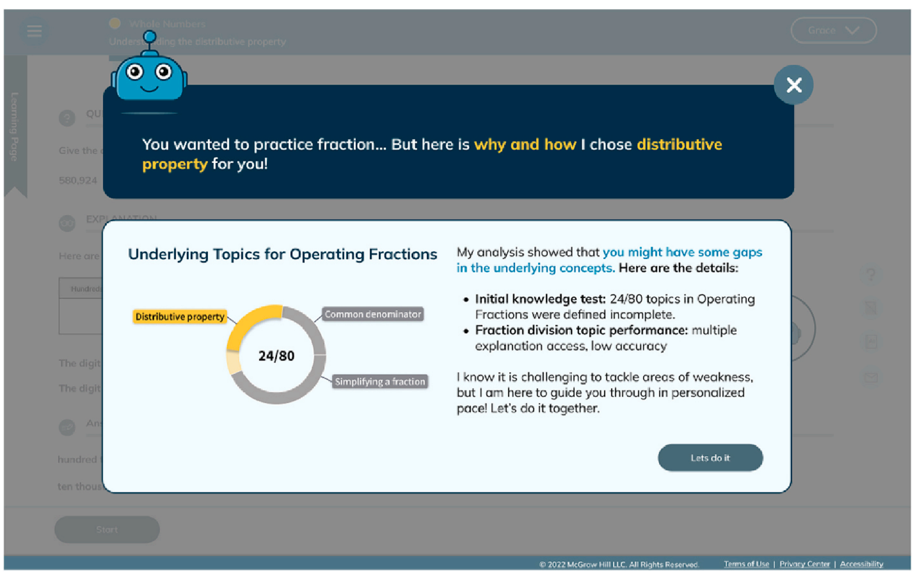

Fig. 6. Explainable AI feature: Practice question selection details.

why, leading to confusion. This design addresses that issue by showing why and how specific questions are selected, upholding the values of transparency, trust, and helpfulness (Cheng & Fleischmann, 2010; Floridi et al., 2020; Friedman et al., 2005). The design was further enhanced with feedback from a UX designer for better visual explanations.

Next, Fig. 7 highlights a human-in-the-loop feature. Students frequently expressed frustration with ITS's automated learning paths, which often limited their review opportunities after passing a topic, even when they felt unprepared to move on. This design allows students to communicate their confidence level in mastering a topic, enabling more control over future review opportunities. It emerged from insights gathered in Step 6 and aimed to upholds value of ownership, autonomy, accountability, and self-respect (Cheng & Fleischmann, 2010; Friedman et al., 2005).

Fig. 8 presents another human-in-the-loop feature, allowing students to communicate whether they received external help. Both students and instructors noted that while ITS assumed correct answers reflect mastery, many students rely on external resources, leading to discrepancies in personalized learning paths. This design lets students inform ITS when they have used outside help, thereby improving the accuracy of their learning paths and tailoring future support to individual preferences. This was also derived from Step 6 insights and aimed at upholding values of ownership, autonomy, accountability, and self-respect (Cheng & Fleischmann, 2010; Friedman et al., 2005).

Finally, Fig. 9 showcases a privacy control feature. Initially, students lacked awareness of privacy issues in edtech, but once informed, many became curious about how their learning data was being used. This design enables students to learn about data repurposing and offers control over how their data is used, following advice from an AI ethics researcher to give users more choices regarding privacy. The goal of this design was to uphold values of privacy, informed consent, and self-respect (Friedman et al., 2005).

5. Discussion

This section outlines implications of applying VSD in education and student-centered AI principles in IT, limitations, and suggested

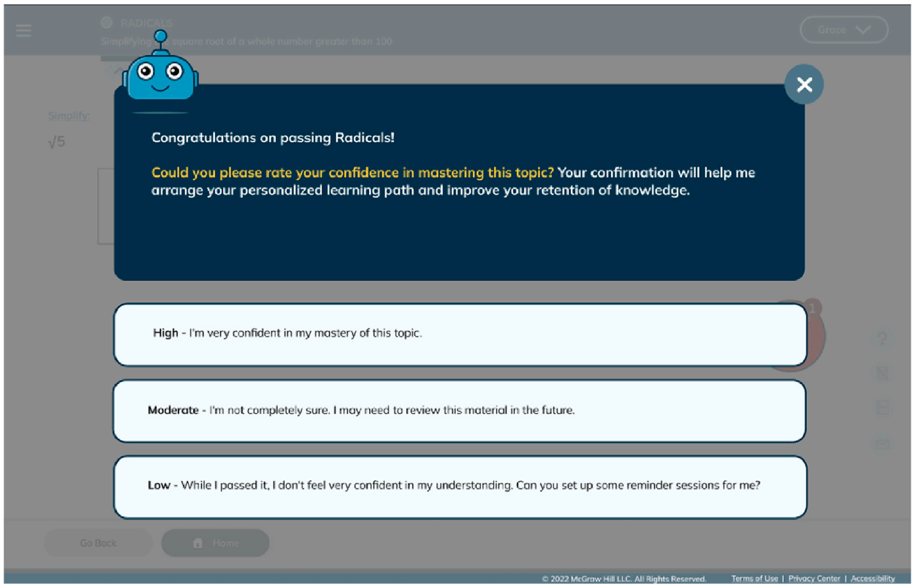

Fig. 7. Human-in-the-loop feature: Confirming students' confidence level over topic mastery.

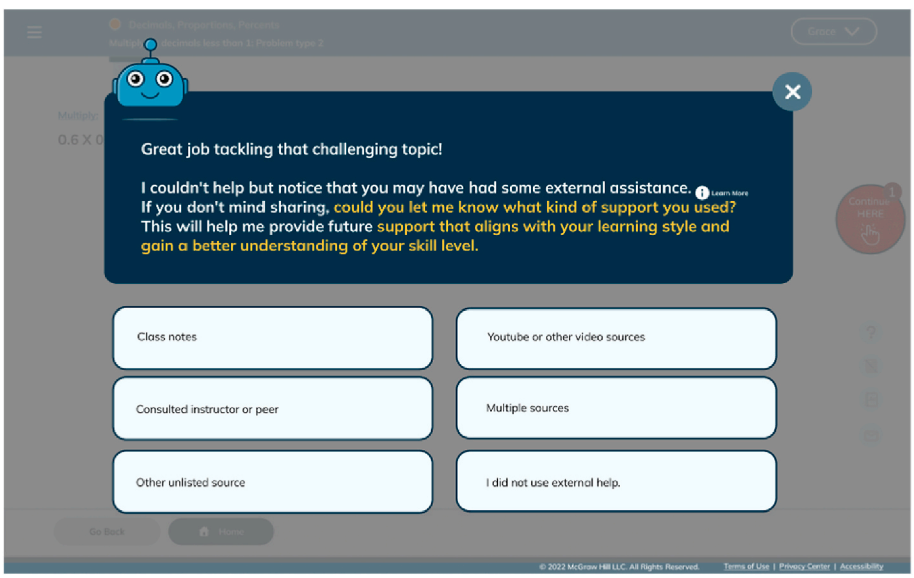

Fig. 8. Human-in-the-loop feature: Communicating receiving external help.

directions for future research.

5.1. Implications for applying VSD in education

Despite 80% of American students utilizing learning platforms daily (Fittes, 2021), they often lack control over how these systems are designed and selected for their learning, suggesting historical oversight of student and instructor values in learning platform design. This aligns with (Viberg et al., 2023), who found that designers tend to embed their own values into AI-based learning systems, consciously or not, which can negatively impact students and educators. For instance, the designer's decision to incorporate a learner dashboard may trigger competition among students, which may be against student and teacher's values of fostering a collaborative and supportive learning environment.

Recognizing this gap in the historical oversight of student and instructor values in edtech design, this study engaged students and instructors who have substantial, real-world experience using ITS. Their prolonged ITS usage experience ensured that interviews captured user values, needs, and pain points related to using ITS. This is in line with Barreiros, Leitner, Ebner, Veas, and Lindstaedt (2023) who emphasized the value of including educational stakeholders in participatory research to explore students' needs and co-create solutions. However, they noted that participants in such studies are often high-performing students and engaging with lower-performing students is challenging. In contrast, this study worked students enrolled in developmental math courses at community colleges, who are generally not high performers in math. Many participants were returning to community college after significant gap and some struggled to pass the course due to a lack of understanding of ITS. Also, their academic success was directly tied to their performance on the ITS, as it determined their passing of the course. By engaging with this demographic, the study aimed to reflect the values and lived experiences associated with ITS use.

With the growing interest in AI ethics in education, many researchers have advocated for VSD to investigate ethical and human-centered AI-based learning platform designs (Chen & Zhu, 2019; Li & Gu, 2023;

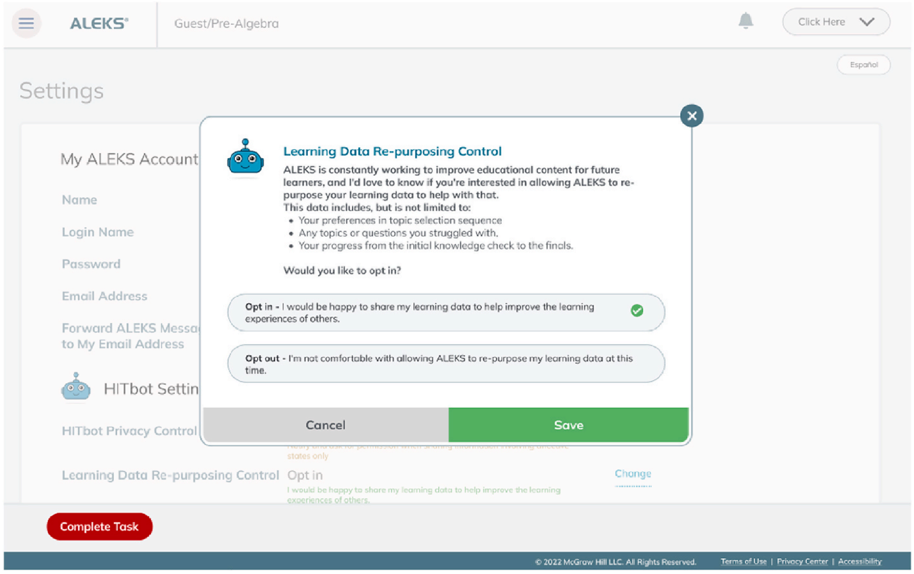

Fig. 9. Privacy control feature: Learning data repurposing control.

Prieto et al., 2023, 2024; Viberg et al., 2023; Zhu et al., 2018). Regardless of the interest, its current application in educational technologies remains limited (Prieto et al., 2024). A recent review of 54 VSD publications discovered that only one VSD study applied to education (Gerdes & Frandsen, 2023). By utilizing VSD, this study showed how student perspectives, values, and needs can be integrated into ITS (Viberg et al., 2023).

In this study, VSD structured the inquiry into three distinct yet connected lenses through conceptual, empirical and technical investigations. The VSD process revealed important gaps and revisions between what we anticipated conceptually, what we observed empirically, and what was feasible or meaningful technically. Conceptually, for example, we treated transparency and model-level confidence as straightforwardly desirable, consistent with AI ethics guidelines. Empirically, however, students and instructors found raw confidence indicators confusing. Technically, these findings pushed us to move away from exposing internal model parameters and toward human-centered reframing, which better aligned with both learner goals and instructors' practices. A similar pattern emerged around privacy and agency: conceptually we framed privacy and opt-out as ethical requirements; empirically we saw tensions in students' privacy preferences and instructors’ concerns about losing insight; technically we had to design controls that respected student choice without completely undermining instructional support. These comparisons show that VSD in ITS does not simply apply values at one level, but instead uses the differences between conceptual, empirical, and technical lenses to surface where values collide, where assumptions break, and how designs must be adapted for educational AI to be both ethically grounded and pedagogically and technically workable.

Therefore, in our study, the three components of VSD showed complementary multi-layered perspective, an approach needed in settings where design decisions may quietly privilege system efficiency or institutional goals over learner values. However, the contribution is not a universal template for ITS design, but a demonstration of how VSD can be used for developing context-specific, value-explicit adaptations tailored to the unique needs of each educational setting. This approach aligns with (Holmes et al., 2021), who emphasized creating school-based intelligent resources that cater to the specific needs of different regions and schools. Through VSD, we developed preliminary design concepts that incorporate trust-building, explainable AI elements tailored for community college students, considering the unique educational and demographic backgrounds of community college developmental courses. This meticulous consideration of the target audience and their technological interactions underscored the value of conducting systematic VSD processes. By conducting conceptual, empirical, and technical investigations, educational technologies can be designed to genuinely serve students and provide them with context-specific and meaningful learning experiences.

5.2. Implications for applying responsible AI in ITS

In this study, the themes of trust, agency, and privacy emerged from the interaction of VSD's conceptual, empirical, and technical investigations. Conceptually, VSD guided us to identify these values as central focus within an ITS context. Empirically, value-oriented interviews with students and instructors revealed how these tensions were actually lived or perceived. For example, raw model-confidence displays that seemed conceptually aligned with transparency did not build trust in practice, while limited review control constrained students' sense of agency. Technically, the VSD process made these conflicts actionable by enabling us to test concrete alternatives, such as human-centered explainable-AI screens that foreground why a question is given, or human-in-the-loop controls that support bounded learner autonomy. Through this process, VSD did not simply surface the values of trust, agency, and privacy; it allowed us to experiment with how they could be realized, interpreted, resisted, or misunderstood in interaction, and to refine features when value alignment broke down. This iterative testing makes VSD a meaningful approach for shaping responsible AI in educational contexts. Building on these VSD-derived insights, this section outlines key design implications for responsible AI in ITS, particularly through explainable AI, human-in-the-loop features, and privacy controls.

Trust is a foundational value influencing both learners and instructors' willingness to let ITS make decisions and collaborate toward optimal learning outcomes (Nazaretsky et al., 2022). Transparency is a critical component of trust since clearly conveying educational information fosters openness and trust in ITS (Li & Gu, 2023). In their comprehensive analysis of human-centered AI factors, Li and Gu (2023) identified transparency as building blocks for credible AIED systems. This aligns with the present study's emphasis on designing explainable AI (XAI) features to enhance transparency grounded in human-centered AI principles. In this study, through XAI, the prototype communicated the educational principles behind ITS to earn students' trust while guiding them toward beneficial learning paths that respect their choices.

In the past, enhancing trust between ITS and students was typically the instructors' job, but the inconsistency in between different instructors' efforts were problematic (Viberg et al., 2023). By integrating these XAI features, our study shows how ITS can inherently deliver the educational values to students building trust towards it.

Agency, defined as "the extent to which students have choices about what to do and when and how to do it" (Schunk, 2012), is a critical component for responsible learning technologies. When developing responsible ITS, it is crucial to consider ways to involve students in decision-making processes, thereby providing them with opportunities to develop agency and self-regulation. Researchers (Li & Gu, 2023; Viberg et al., 2023) emphasize developing autonomy through learning analytics systems and the need for innovative models that balance human and machine intelligence in education. Accordingly, this study aimed to address student-ITS collaboration by incorporating human-in-the-loop features enabling student agency. Specifically, it allowed students to indicate their confidence in mastering a concept instead of solely relying on AI decisions. Students could therefore control their review opportunities together with the ITS, strengthening their metacognitive understanding and control over their learning. Moreover, the human-in-the-loop feature was integrated to help students seek personalized assistance when facing challenges. Often, students who are stuck in a concept do not realize they need help or do not know what kind of help to seek. We designed the ITS to automatically prompt students to seek help and choose personalized methods to overcome obstacles. This design effort enhanced student-ITS communication, enhancing student agency in learning with ITS.

The importance of privacy as a core component in ethical AI systems has been extensively emphasized (Floridi, 2019; Holmes et al., 2021). This applies to education as educational researchers (Akgun & Greenhow, 2021; Li & Gu, 2023) highlight the critical role of privacy in designing ethical AI learning systems at both individual and collective levels. Accordingly, this study highlighted the necessity of increasing students' awareness of their privacy and the handling of their learning data in using ITS at individual level. We advocated for the design of ITS to empower students with the ability to opt-out of data sharing and ensure transparency about how their data is utilized by educational institutions and edtech companies. Through these design efforts, ITS can provide effective privacy controls for students, thereby enhancing respect for their privacy and reducing perceived risks associated with using AI in educational software (Li & Gu, 2023).

By embedding these values of trust, agency, and privacy into ITS prototype, we suggest potential directions for developing more ethical and student-centric educational technologies in similar settings.

5.3. Limitations and future work

This study is based on a small and demographically narrow sample from a single institution. While participants' lived experiences provide valuable, context-specific insights, the findings should be interpreted as exploratory and situated rather than broadly generalizable. Future research should include more diverse participants across institutions and contexts to examine how stakeholder values and experiences with AI-based learning systems vary to clarify how cultural and institutional settings shape ethical and pedagogical priorities AI educational system design.

Although the prototype designs developed in this study offer promising directions, they remain low-fidelity and untested in authentic classroom environments, meaning the implications here should be interpreted as situated design insights rather than demonstrated impact. Because our scope and resources limited the development of a fully functional ITS, expert consultations served as an interim way to assess technical feasibility. Future work should translate these conceptual prototypes into full implementations and pilot them in real classroom settings to determine whether VSD-informed design choices meaningfully influence trust, agency, privacy perceptions, or learning outcomes, and to provide the empirical evidence needed to evaluate their practical effectiveness compared with traditional or existing AI-enhanced systems.

Additionally, this study explored only a selection of student-centered AI design techniques, leaving a vast array of other potential techniques unexamined. Future research should explore a broader range of student-centered AI design techniques. This could include techniques such as culturally responsive learning analytics systems (Viberg et al., 2023). By examining a wider variety of techniques, researchers can develop more personalized, inclusive, and equitable educational technologies that address a broader spectrum of learner needs and preferences.

Finally, the current research did not explore the integration of generative AI (GenAI) into the ITS. The core insight from this study is that a GenAI-based tutor chatbot should not be a mere add-on but carefully integrated with ITS to best engage students and respect their individual values. Therefore, building on this study's insights, future studies could explore how and when GenAI should support students in interacting with ITS, especially involving students as a core collaborator in shaping personalized learning paths. Specifically, future studies should investigate how GenAI can be utilized to enhance the student-centeredness of ITS. This could involve employing generative AI to create more dynamic and responsive learning content, generating personalized learning pathways, and simulating human-like interactions utilizing multimodal AI for more engaging educational experiences. By integrating generative AI, ITS could offer richer, more personalized learning experiences that adapt to individual learning styles and needs.

6. Conclusion

The current study explored the design of student-centered ITS. By identifying designs that incorporate human-in-the-loop, explainable AI, and privacy controls, this study provided concrete example of applying human-centricity in ITS. As a few of the studies applied VSD in education (Prieto et al., 2023; Viberg et al., 2023), the study hopes to have demonstrated VSD's contribution in creating value-led edtech products.

CRediT authorship contribution statement

Eunhye Grace Ko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joan E. Hughes:** Writing – review & editing, Conceptualization.

Informed consent

- Consent to participate: Informed consent was obtained from all individual participants included in the study. Online informed consent was obtained prior to the surveys and interviews.
- Consent to publish: Patients signed informed consent regarding publishing their data and all data has been converted to the confidential and anonymous format.

Research involving human participants and/or animals

- The questionnaire and methodology for this study were approved by the Human Research Ethics committee of the University of Texas at Austin (Ethics approval number: STUDY00002943, PI: Kenneth Fleischmann).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT in order to check for grammar and expressions. After using this tool/service, the author(s) reviewed and edited the content as needed and take

(s) full responsibility for the content of the published article.

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Financial interest

- The authors have no relevant financial or non-financial interests to disclose.
- The authors have no conflicts of interest to declare that are relevant to the content of this article.
- All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.
- The authors have no financial or proprietary interests in any material discussed in this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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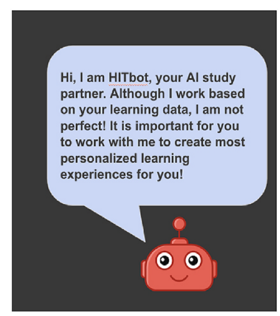
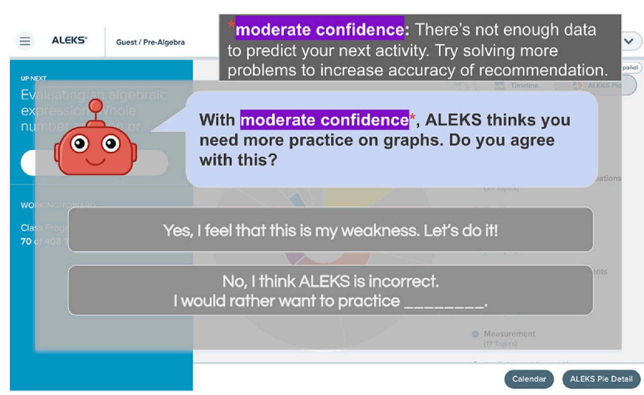
Appendix A. Initial Design: Low-Fidelity Prototype

	Stage	Screen
1	Introduction	
2	Explainable AI: Revealing confidence level	

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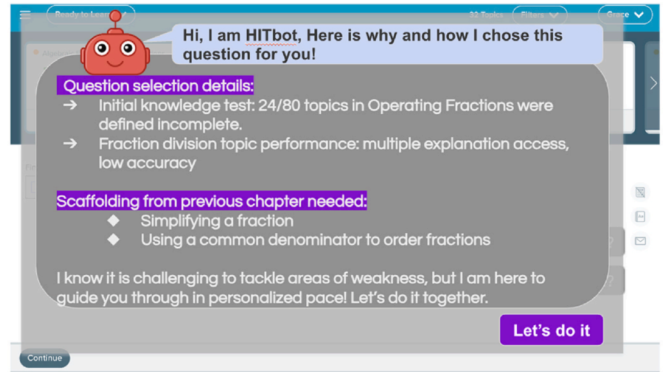
<table>
<tr><th></th><th>Stage</th><th>Screen</th></tr>
<tr><td>3</td><td>Explainable AI: Why and how questions are given</td><td>
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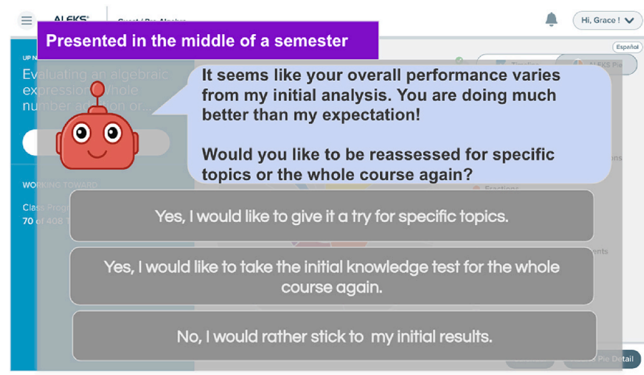
<tr><td>4</td><td>Human-in-the-loop: Retake the Initial knowledge test</td><td>
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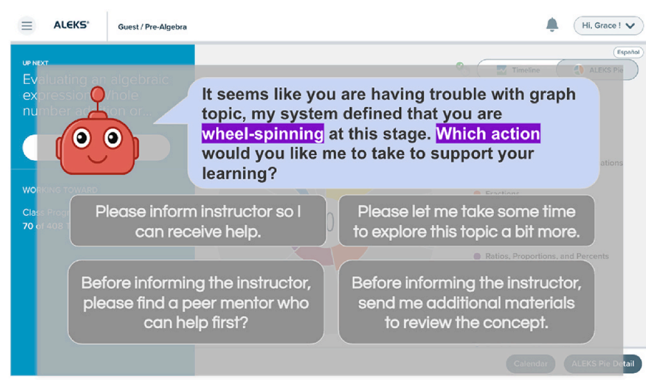
<tr><td>5</td><td>Human-in-the-loop: Asking for student choices in follow-up</td><td>
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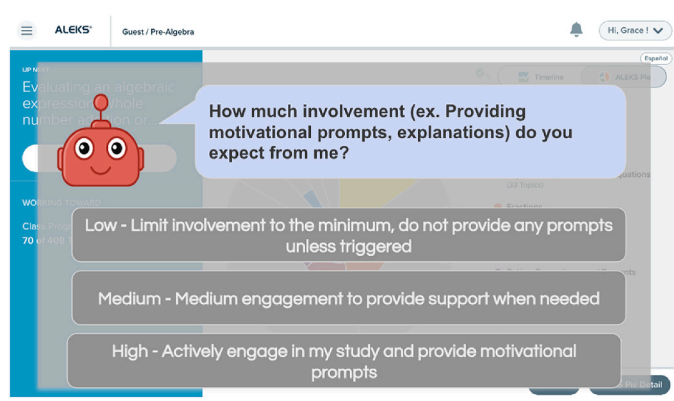
<tr><td>6</td><td>Human-in-the-loop: Involvement level control</td><td>
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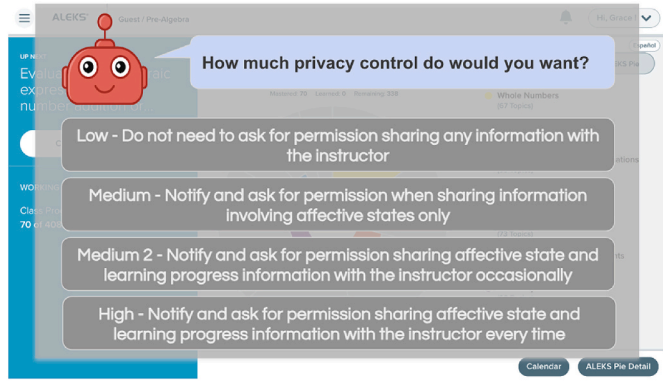
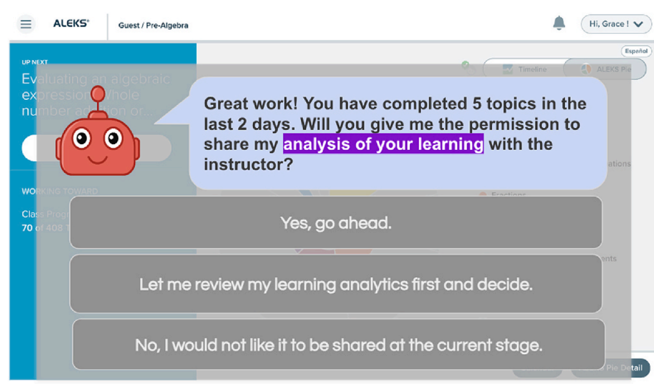
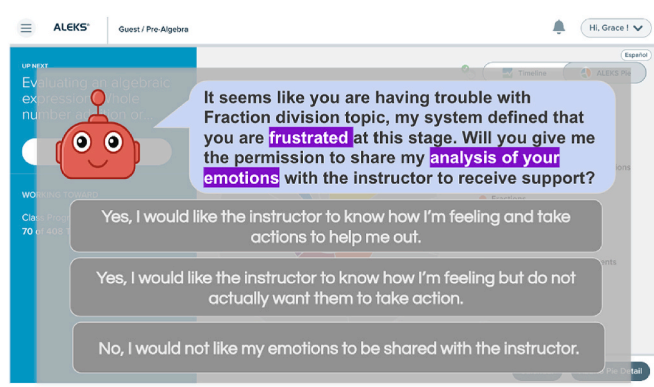
	Stage	Screen
7	Privacy control: privacy level control	
8	Privacy control: Asking permission to share learning Analysis	
9	Privacy control: Asking permission to share affective states	

Appendix B. Step 7. Final Design

Topic/Screen	Screen
Explainable AI Features	

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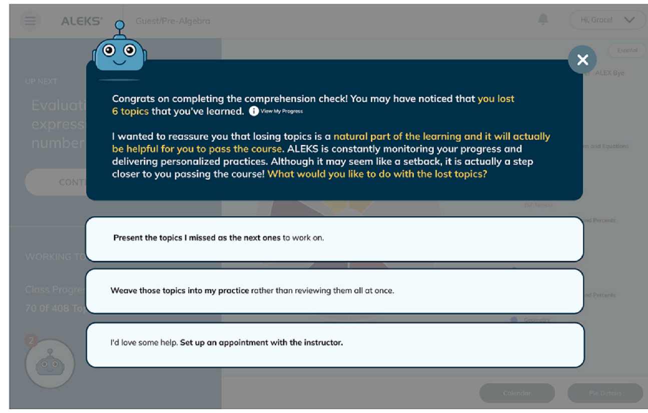
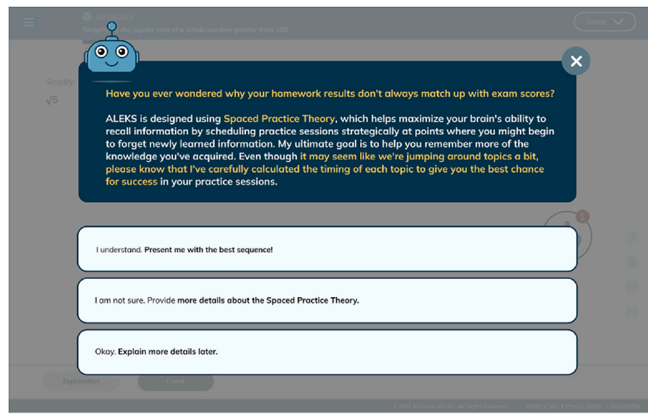
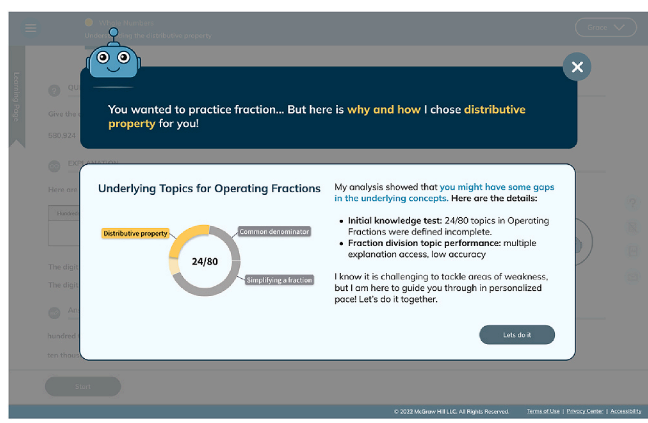
	Topic/Screen	Screen
E1	Comprehension check interpretation	
E2	Setting up students' expectation on learning science behind ITS	
E3	Practice question selection details	

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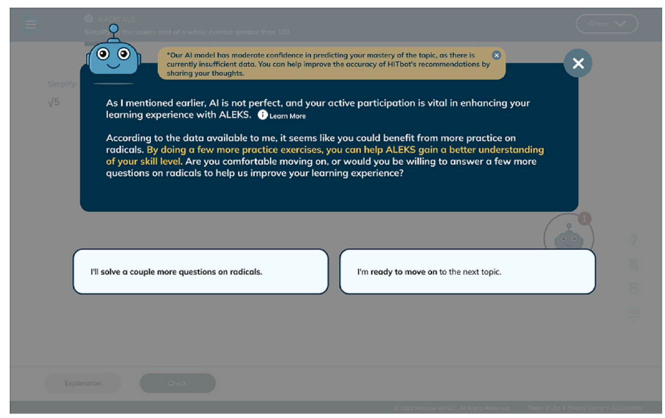
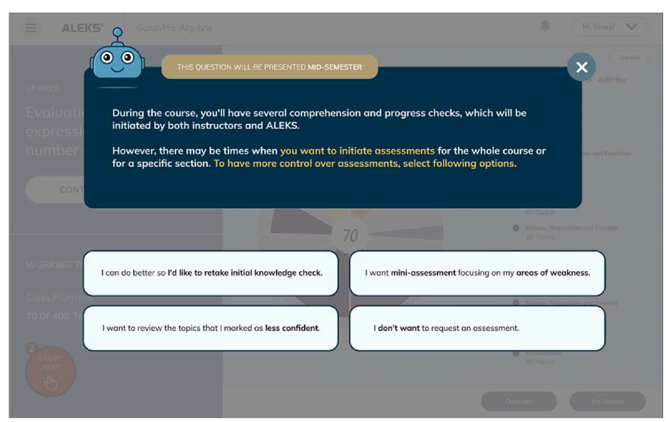
	Topic/Screen	Screen
E4	Concept and learning path connection	
E5	Communicating AI model's confidence level	
Human-in-the-loop Features		
H1	Student control over re-assessment	

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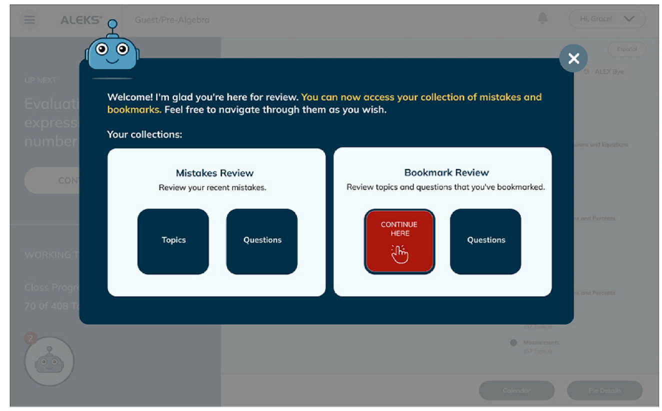
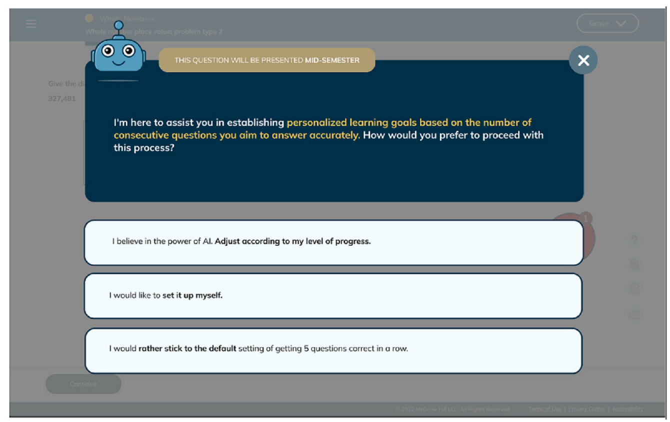
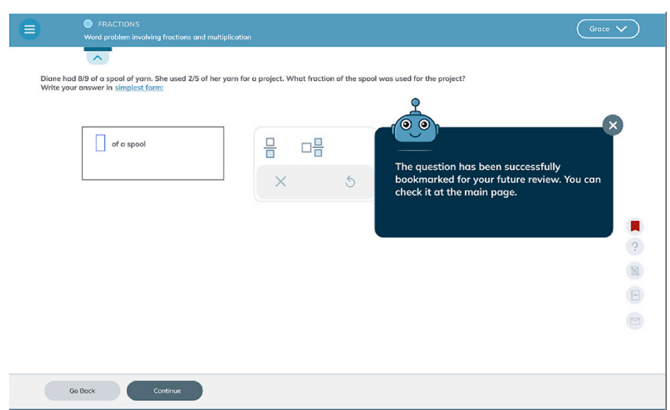
	Topic/Screen	Screen
H2	Student control over review	
H3	Setting up personalized learning goals/pace	
H4	Bookmarking topic/questions for review	

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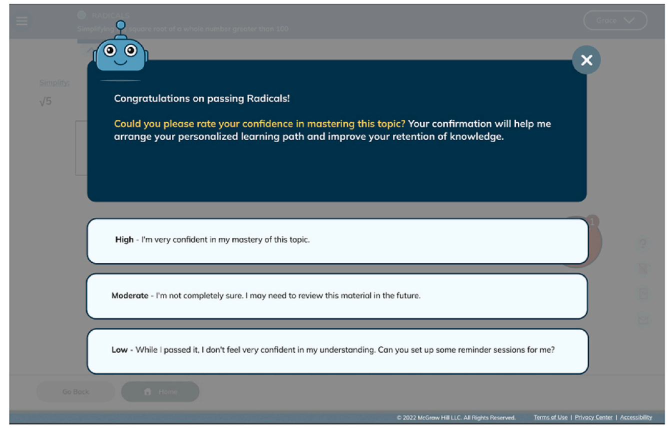
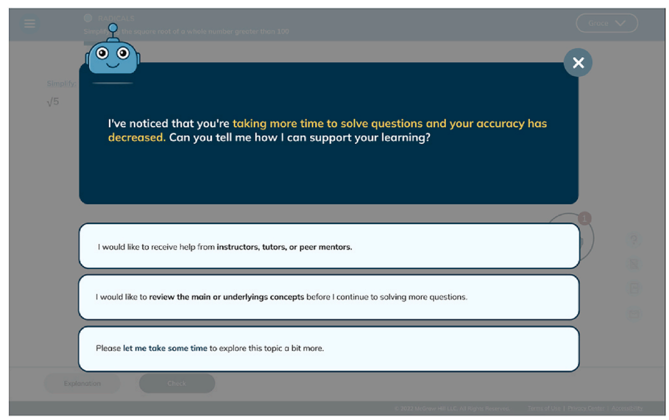
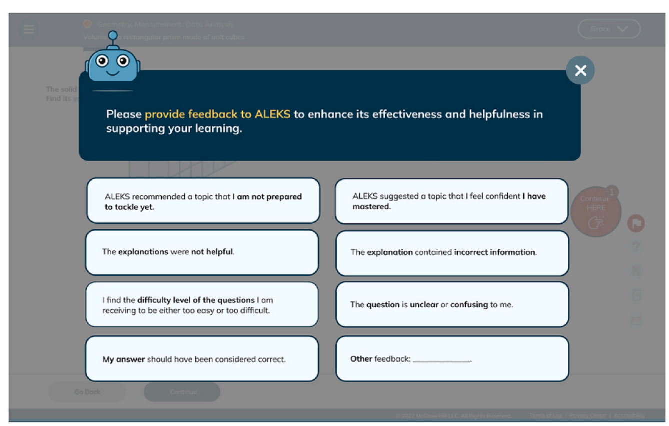
	Topic/Screen	Screen
H5	Confirming students' confidence level over topic mastery	
H6	Personalized support in response to extended answer time and observed declines in accuracy	
H7	Providing feedback to ITS to enhance its effectiveness and helpfulness in supporting learning	

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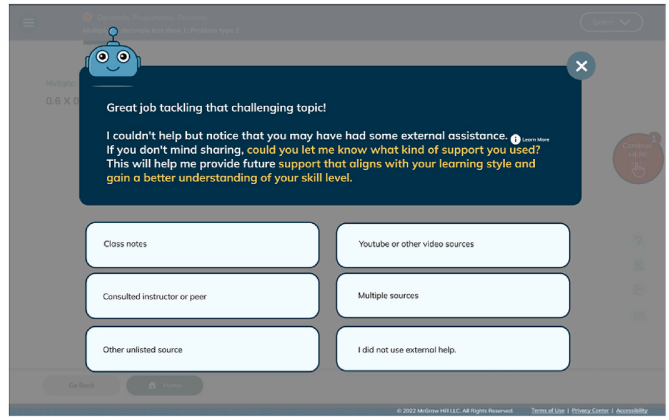
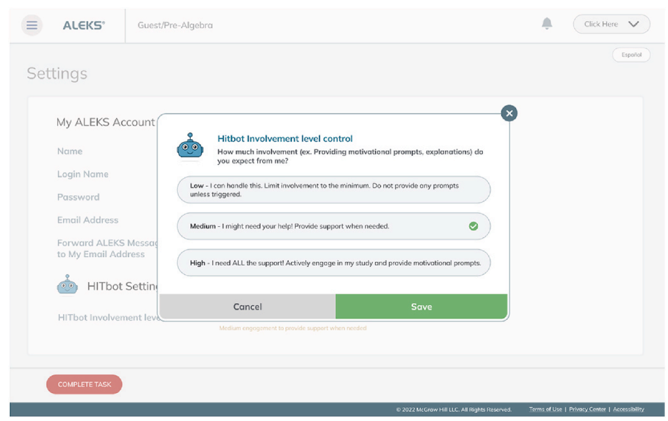
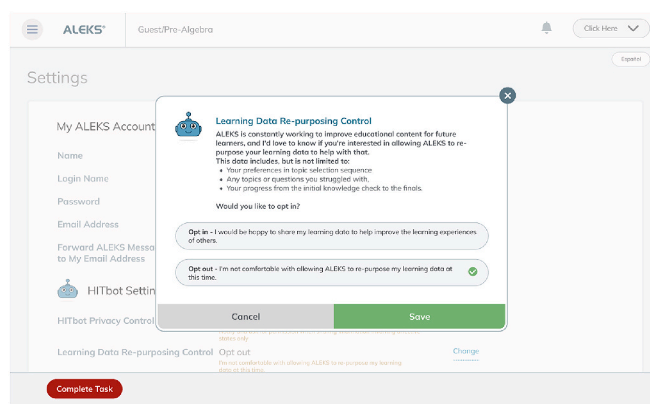
	Topic/Screen	Screen
H8	Communicating receiving external help	
H9	AI assistance involvement level control	
Privacy Control Features		
P1	Learning data repurposing control	

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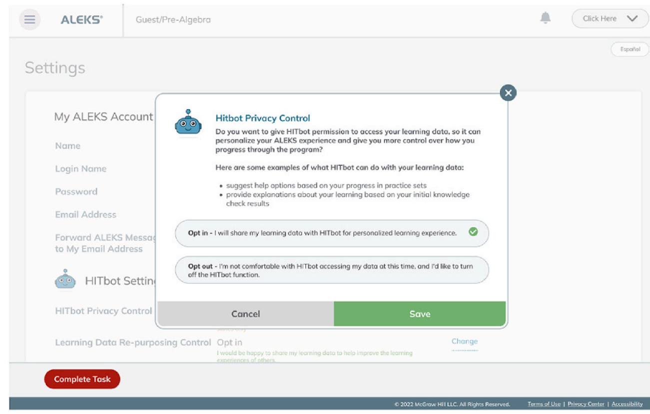
	Topic/Screen	Screen
P2	Privacy control	

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